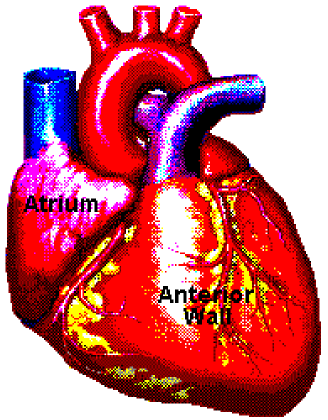
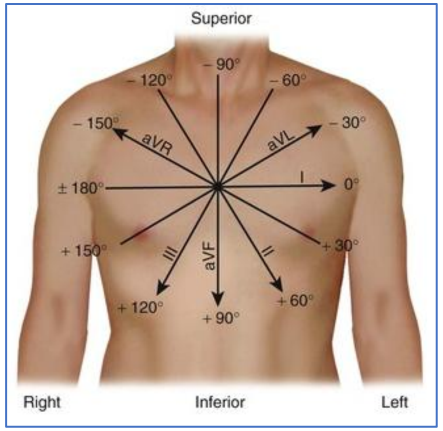


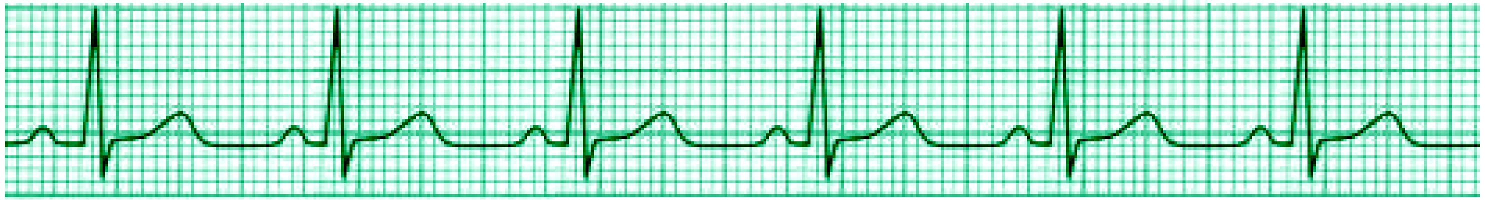
Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 14



ECG Leads, Mean Electrical Axis & Interpretation



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Recommended Readings

Chapter 12: Electrical Activity of the Heart: The Electrocardiogram. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 11: Fundamentals of Electrocardiography: Guyton and Hall Textbook of Medical Physiology, 14th Edition. John E. Hall PhD and Michael E. Hall MD, MS.

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0175 Explain the 12 leads used in standard ECG (I, II, III, aVR, aVL, aVF, V1-V6) including electrode placement, classification according to the plane and polarity, and lead axis.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0176 Explain why the ECG tracing looks different in each of the 12 leads.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0177 Discuss the concept of mean electrical axis of the heart, its normal range and the different methods to determine it.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0178 Describe the alteration in conduction responsible for most common arrhythmias: tachycardia, bradycardia, A-V block, atrial and ventricular fibrillation.

Lecture Outline

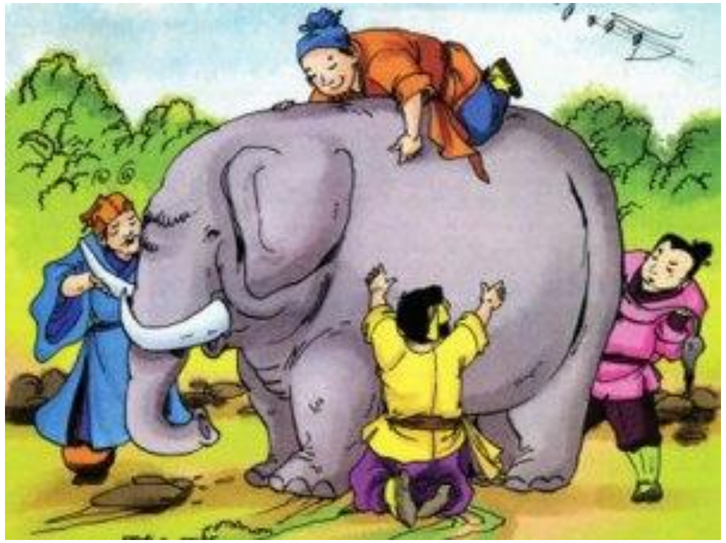
- ECG Leads and placement
- Generation of the hexaxial reference system
- Calculation of the Mean Electrical Axis & its importance
- ECG changes in cardiac arrhythmias

A standard ECG Recording is a 12 Lead Recording

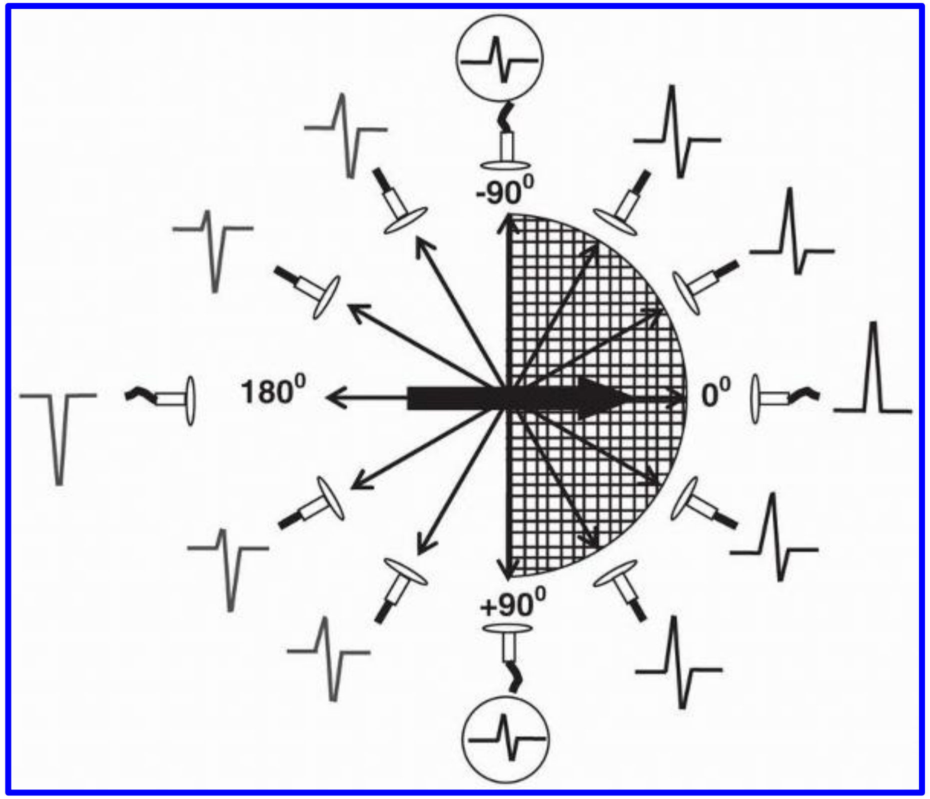
Bipolar Limb Leads	Augmented (Unipolar) Limb Leads	Precordial Chest Leads
Lead I	aVL	V1
Lead II	aVR	V2
Lead III	aVF	V3
		V4
		V5
		V6

A “Lead” is a pair of electrodes (one connected to the +ve input of the ECG machine and the other to the –ve input).

A standard ECG Recording is a 12 Lead Recording: Why 12 Leads?



- Viewing the elephant from many different directions
- Have different views
- Use information from the different men to get the whole picture



12 leads of ECG:

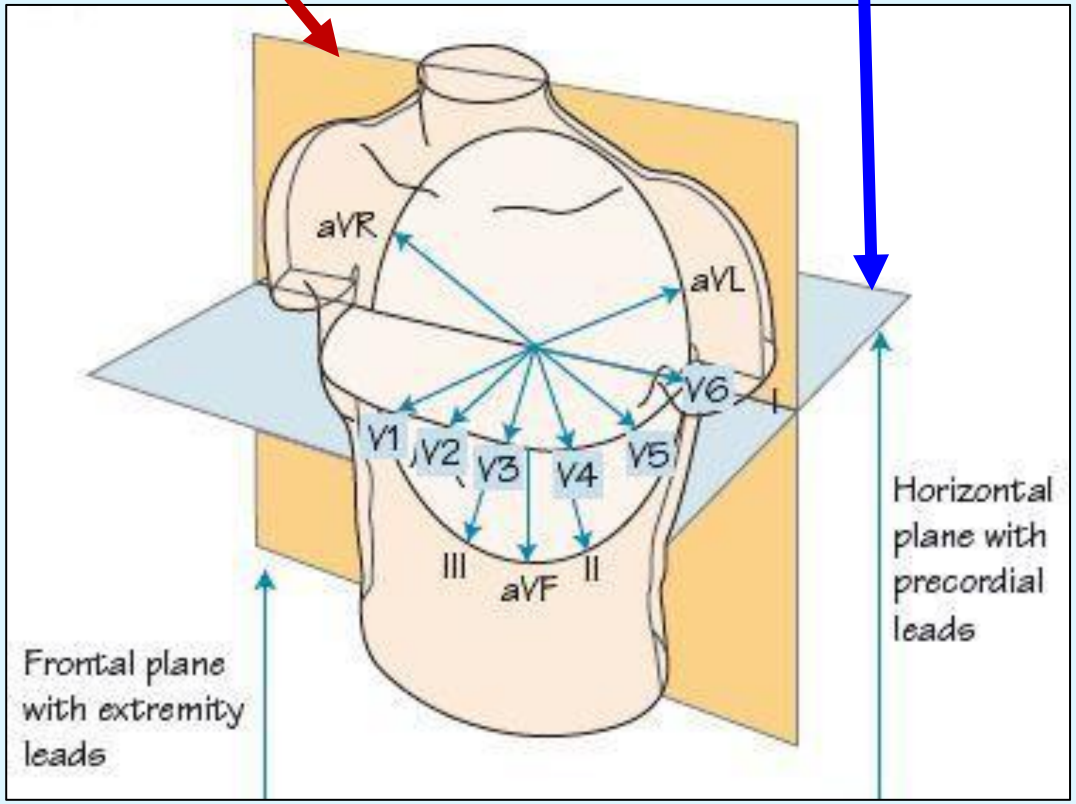
- View the same **QRS depolarization** from different angles through different **leads**
- Each lead gets the whole picture/information about the heart

A standard ECG Recording is a 12 Lead Recording: Why 12 Leads?

- Limb leads view the heart in the **frontal plane**
- Chest leads view the heart in the **transverse (horizontal) plane**
- A single lead will not give all clinically relevant information
- Use of standard 12 leads on body surface gives clinically sufficient relevant information that would have missed otherwise.

Frontal plane
Limb leads

Transverse plane
Chest leads



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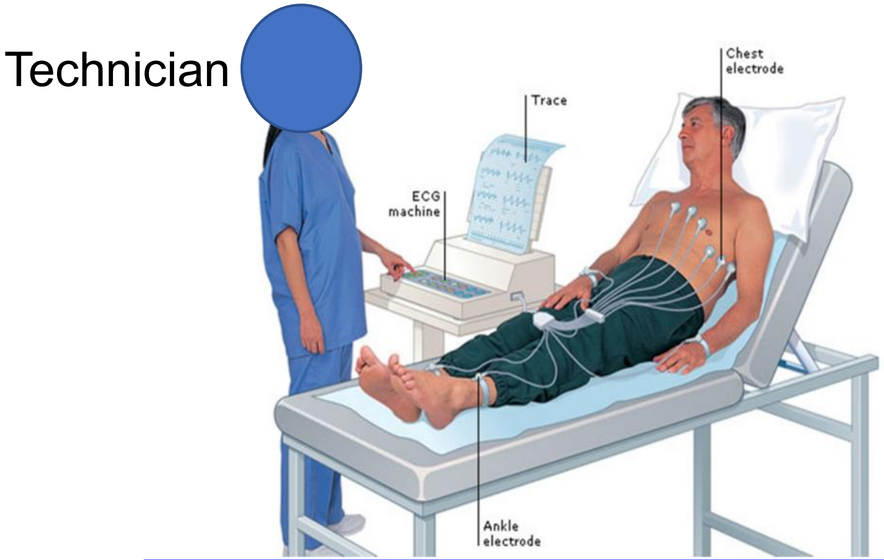
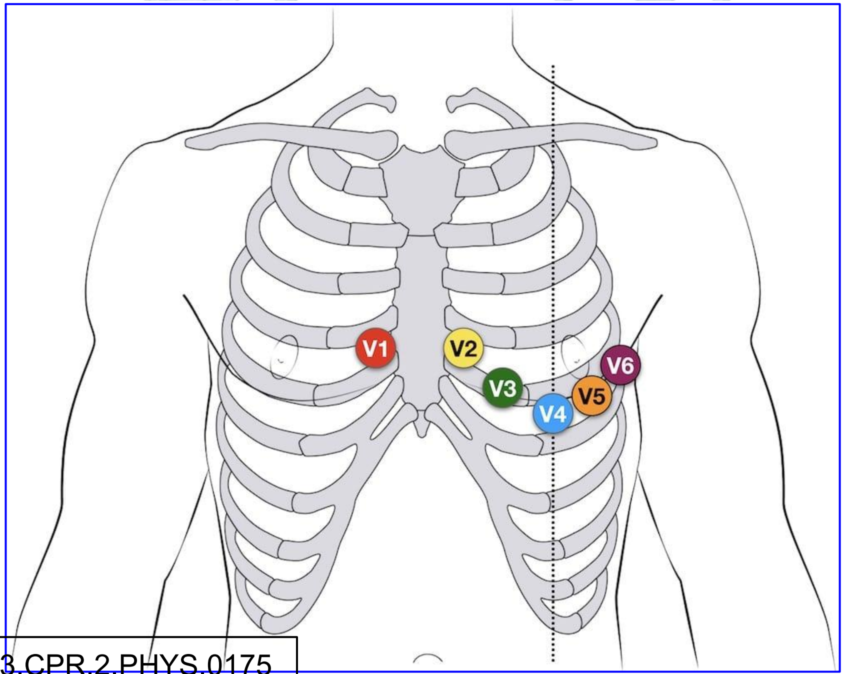
A 12 Lead ECG Recording: Lead Placement



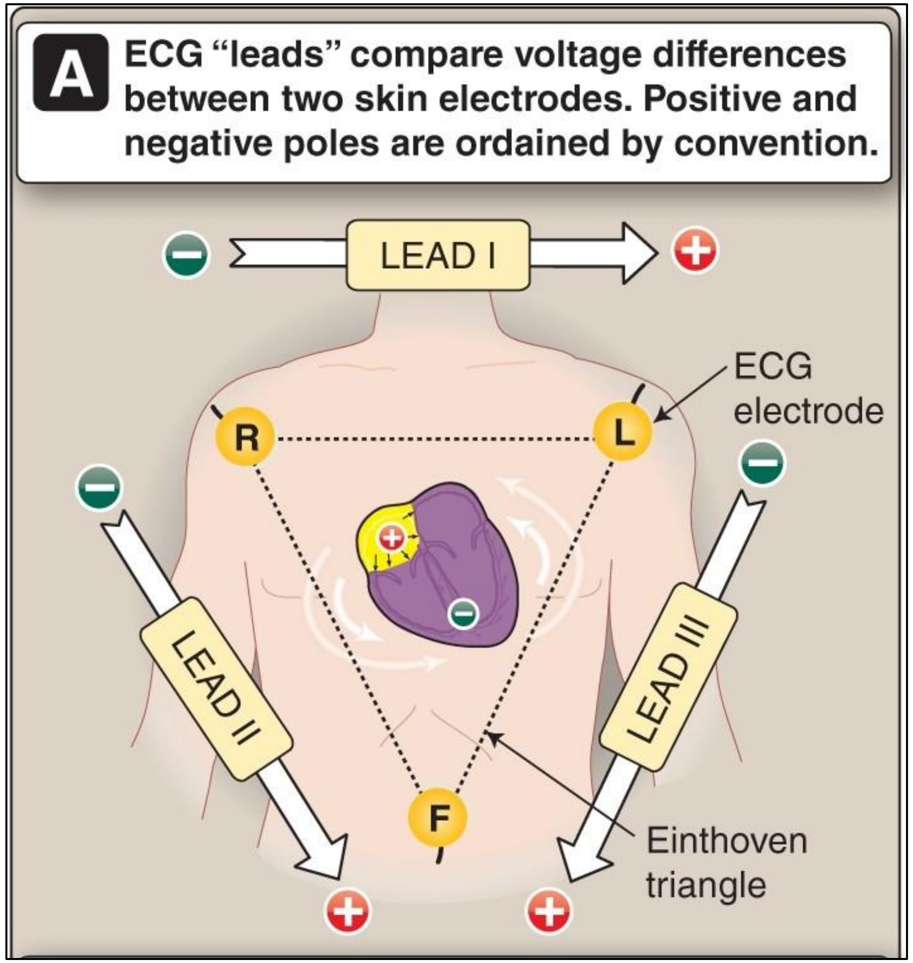
Although the wave of depolarization that goes through the heart is "fixed" the 12 ECG leads record this wave from different angles and directions.

Sites: Right arm, Left arm, left leg and Front of the chest.



Precordial lead placement:
V1: 4th intercostal space (ICS), RIGHT margin of the sternum
V2: 4th ICS along the LEFT margin of the sternum
V4: 5th ICS, mid-clavicular line
V3: midway between V2 and V4
V5: 5th ICS, anterior axillary line (same level as V4)
V6: 5th ICS, mid-axillary line (same level as V4)

Bipolar limb leads: Placement and Polarity

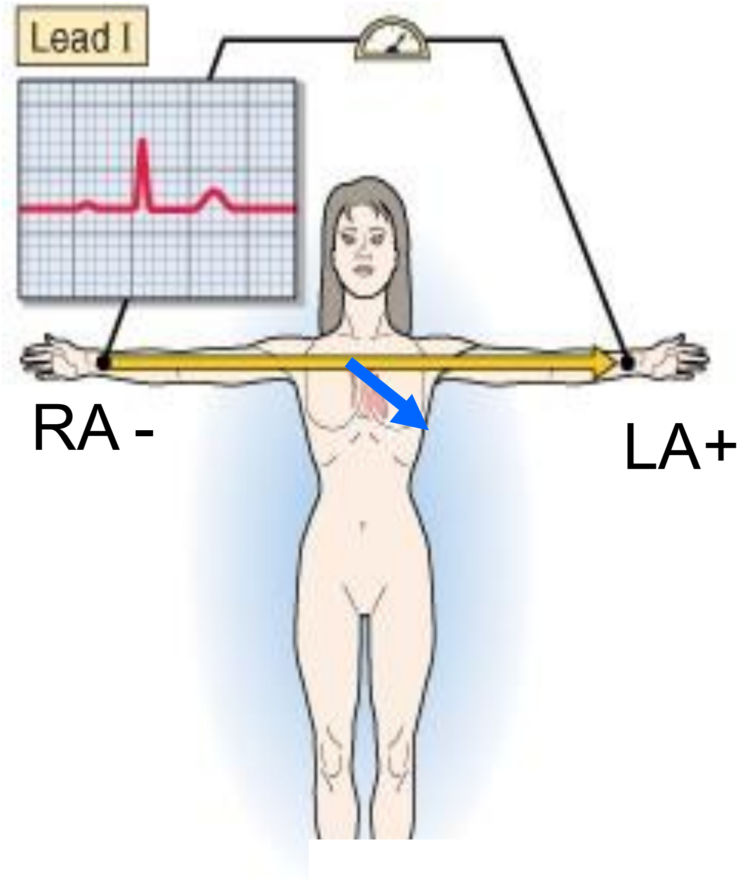


Lead	Polarity of the Lead
I	Right Arm Negative **(-)** and Left Arm Positive **(+)**
II	Right Arm Negative **(-)** and Left Leg Positive **(+)**
III	Left Arm Negative **(-)** and Left Leg Positive **(+)**

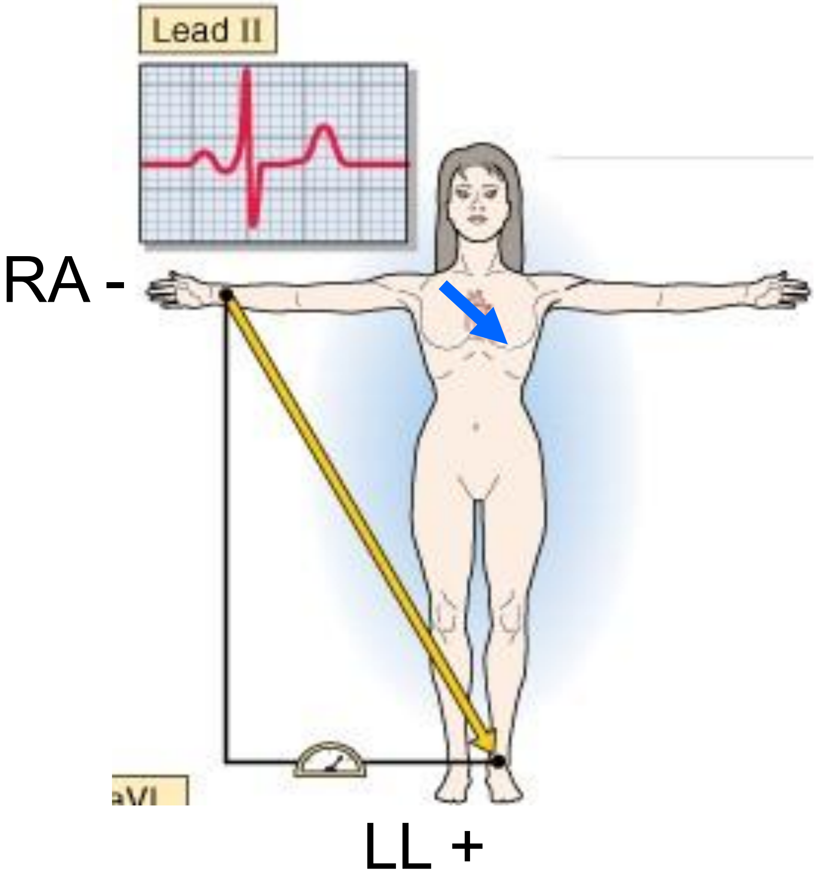
How to remember polarity of these Bipolar leads:
Left Leg is always Positive (LLP);
Right Arm is always Negative (RAN)

Note: arrow heads of Leads denote the +ve electrode

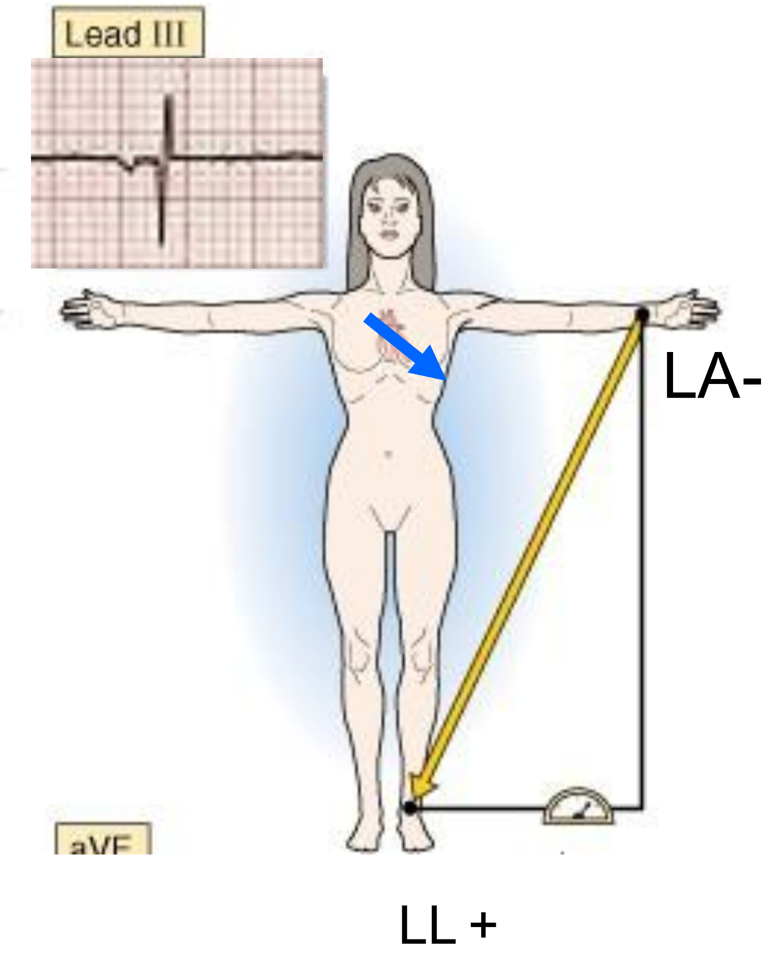
Bipolar limb leads: Lead axis and QRS Vector relationship



Lead I



Lead II

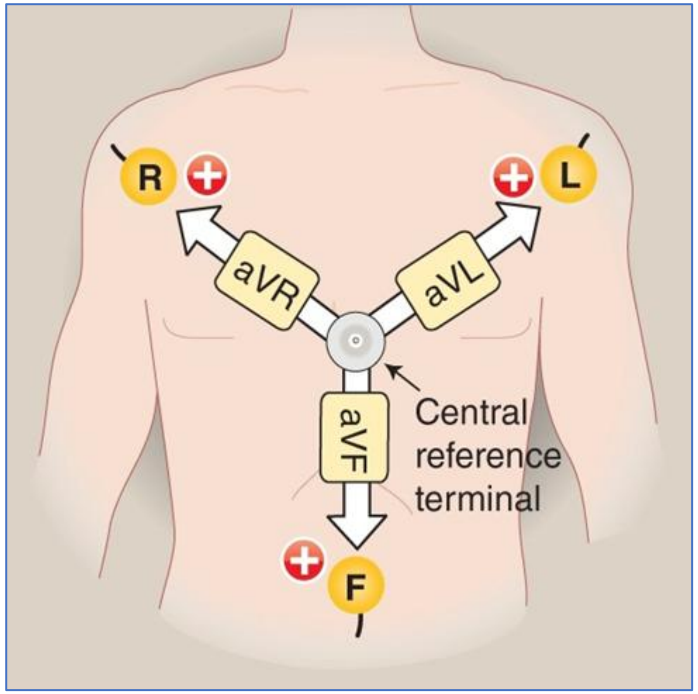


Lead III

Augmented Limb leads: Placement and Polarity

Lead (augmented limb lead)	Polarity of the Lead
aVR (R= right arm)	Right Arm Positive (+) and "V" as Negative (-) electrode
aVL (L= left arm)	Left Arm Positive (+) and "V" as Negative (-) electrode
aVF (F= foot)	Left Foot Positive (+) and "V" as Negative (-) electrode

"V" is an indifferent electrode which is made to have zero potential. This connection is done in the ECG machine.

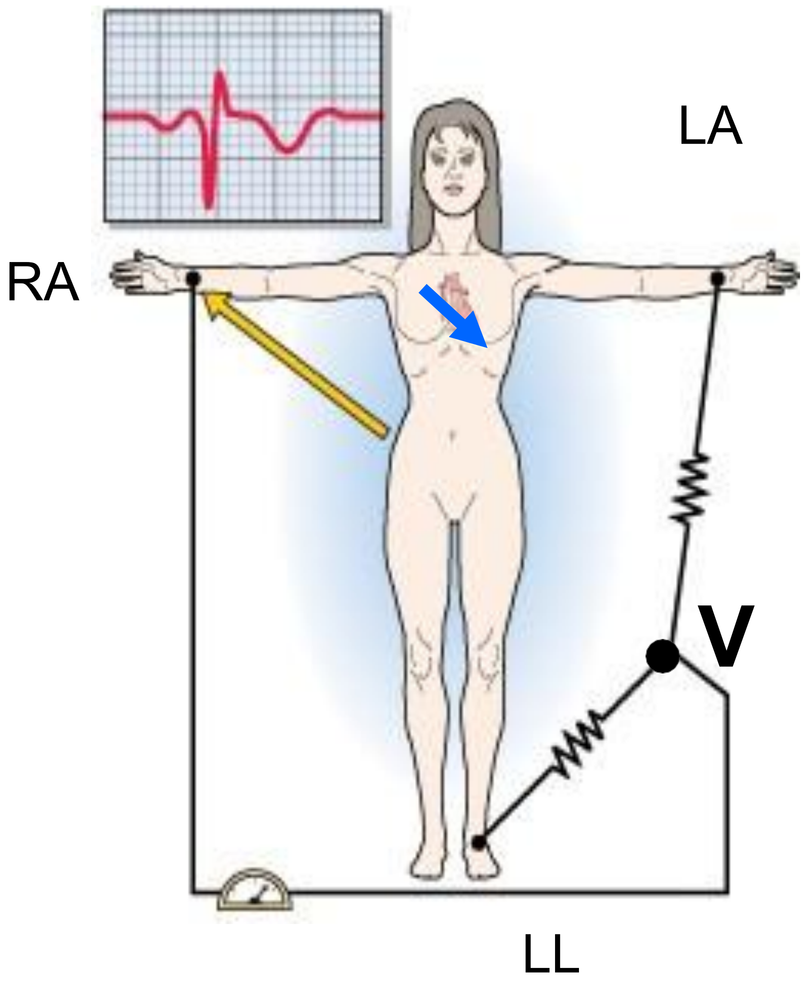


Augmented limb leads measure the voltage difference between an exploring (+ve) electrode placed on one limb (RA, LA or LL) and an indifferent electrode (V) placed at **zero potential**, a central reference terminal (Hence the name "unipolar")

Note: The exploring electrode on the limb is **always** the positive electrode)

Augmented limb leads: Lead axis and QRS Vector relationship

aVR

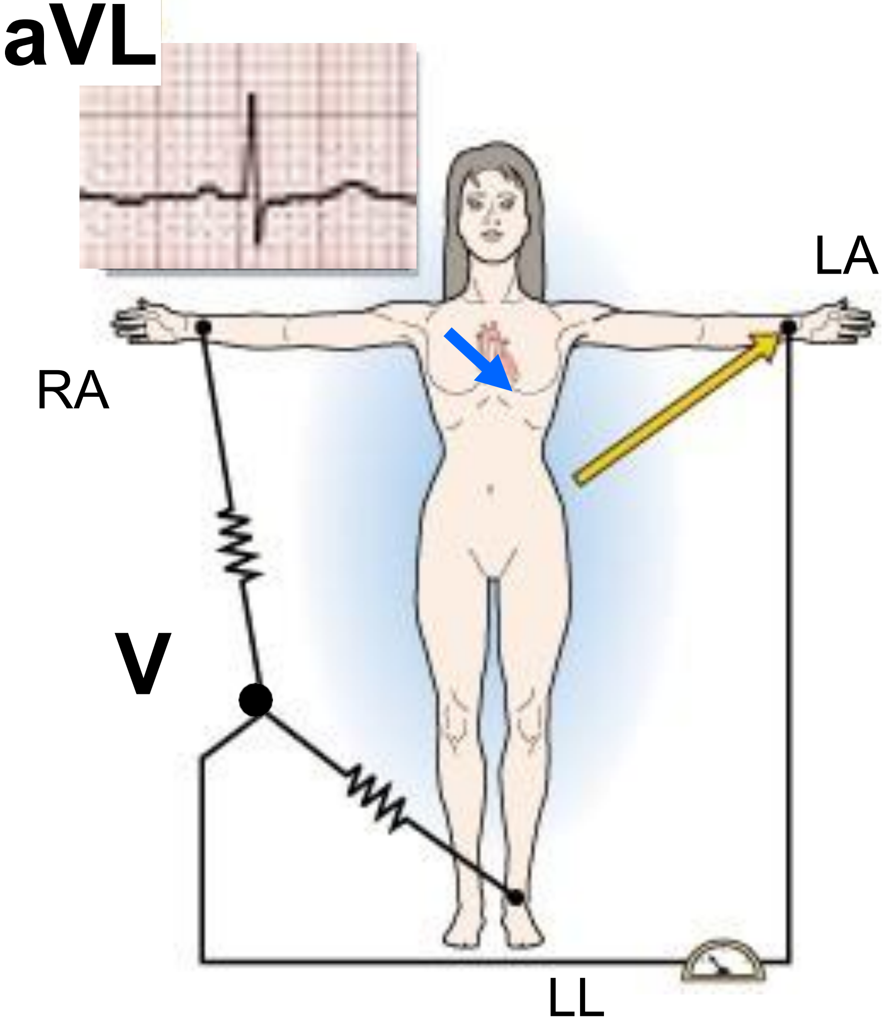


Record the electrical activity between an "exploring electrode" (X) on RA and an indifferent electrode "V". RA is +ve.

The recording axis (yellow arrow) is as shown with the +ve electrode on the right arm)

The mean QRS vector is in the opposite direction to lead aVR, thus QRS complex is largely negative with a relatively large downward deflection.

Augmented limb leads: Lead axis and QRS Vector relationship



Record activity between the +ve "exploring electrode" on LA with an indifferent electrode "V"

The mean QRS vector → is partially in direction of lead aVL, thus QRS complex is biphasic with usually a net upward deflection.

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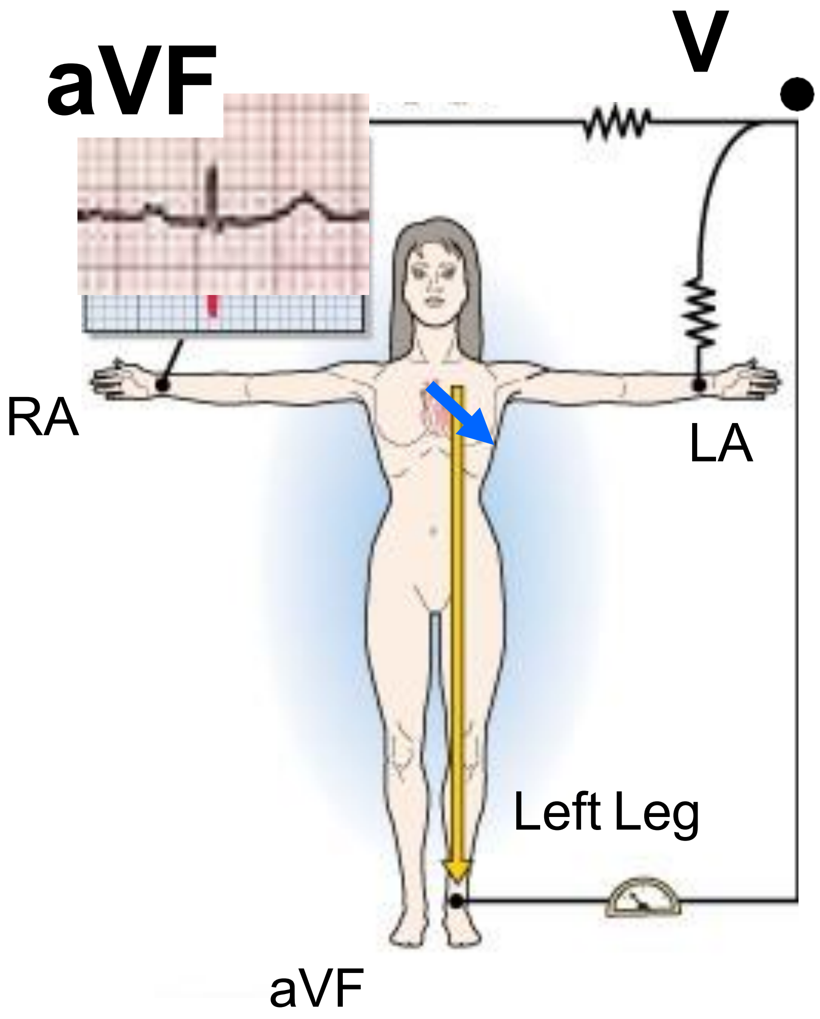
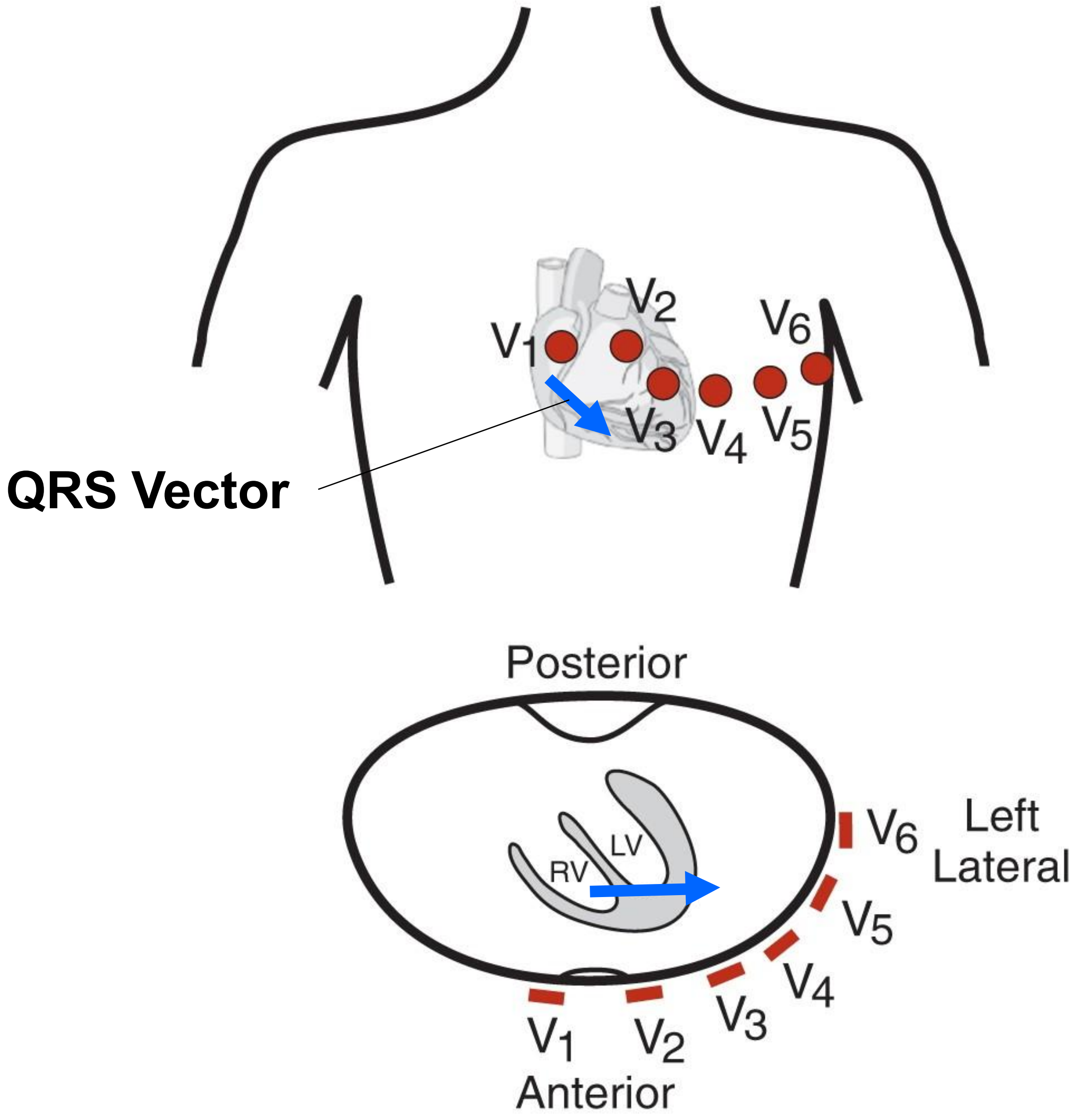
Augmented limb leads: Lead axis and QRS Vector relationship

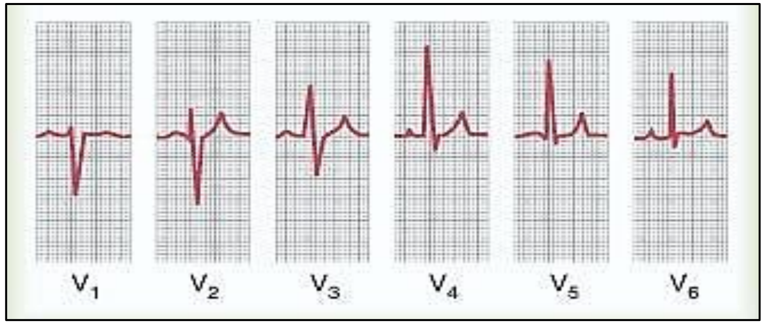


Record electrical activity between the +ve "exploring electrode" (X) on the LL (foot) and an indifferent electrode "V"

The mean QRS vector → is mostly in direction of Lead aVF, thus QRS complex is largely upward.

Precordial Chest Leads: QRS Vector relationship



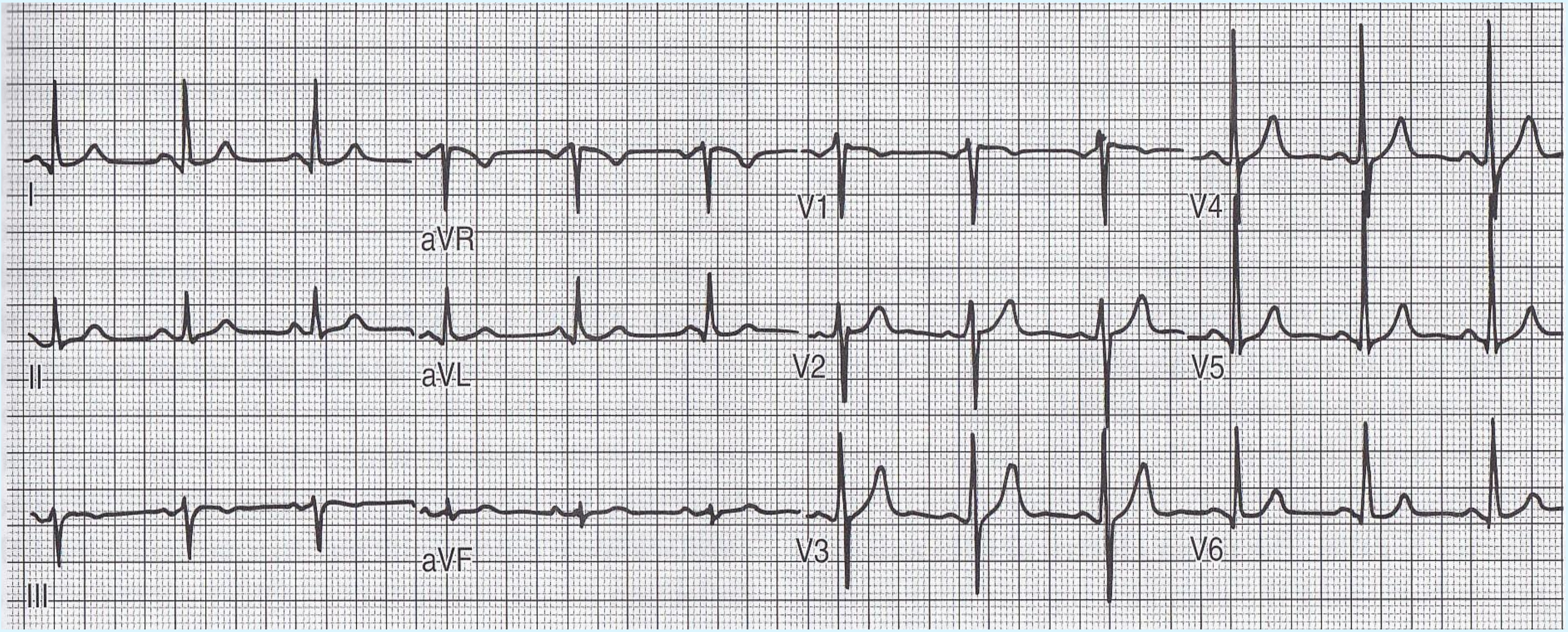


Positive (recording) electrodes are placed on the chest wall and indifferent electrode functions as the negative

$V_{1,2}$ "looks at anteroseptal" (RV, septum)
$V_{3,4}$ "looks at anteroapical" (septum, apex)
$V_{5,6}$ "looks at anterolateral" (LV)

Note change in QRS complex:
$V_{1,2}$: Downward deflection mostly
V_3 : Equiphasic
$V_{4\text{-}6}$: Upward deflection mostly

A Typical 12 Lead ECG Recording



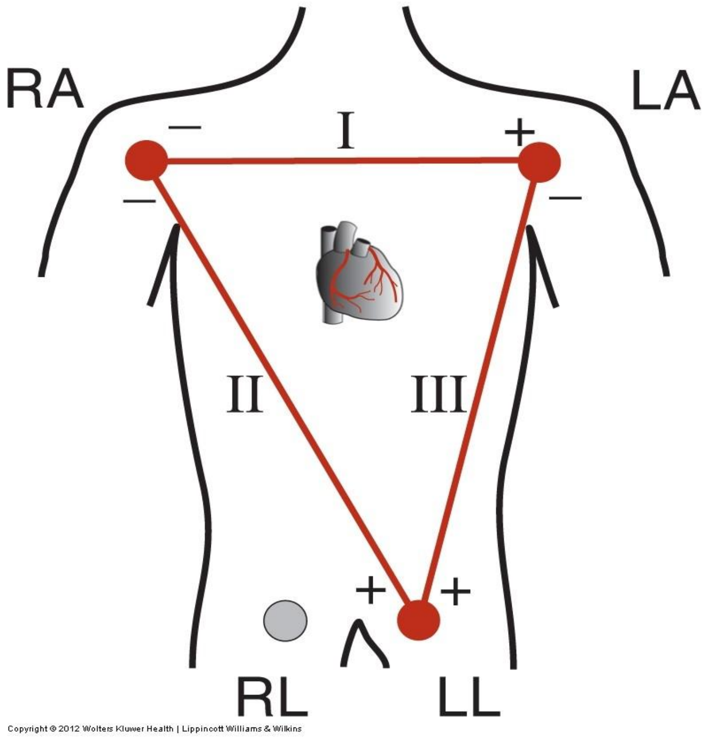
Note the variation in QRS complexes.

QRS in I and II is net positive

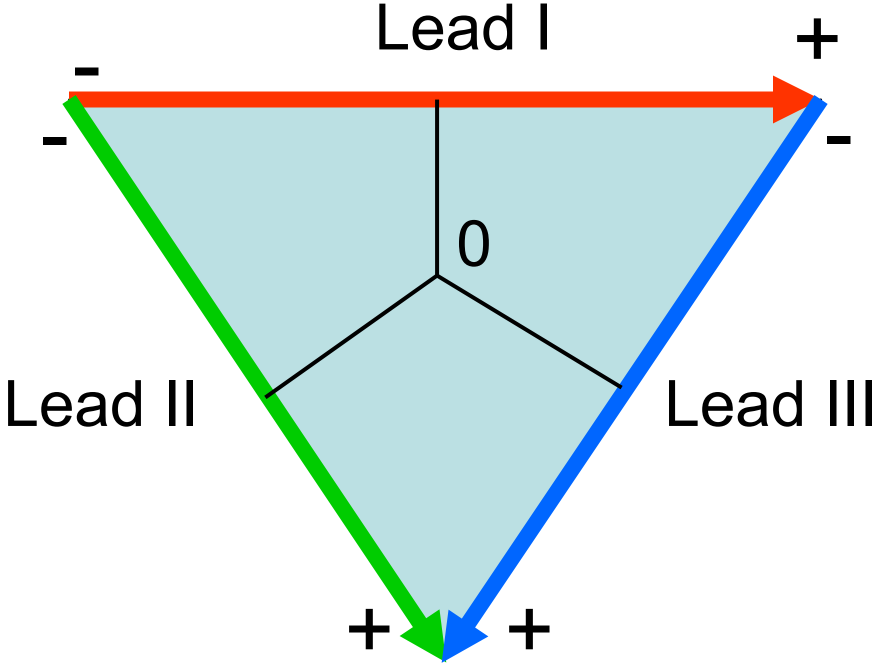
aVR is net negative; aVL is net +ve

Note the polarity changes from V1 to V6

Einthoven's Triangle and Einthoven Law



The 3 pairs of bipolar limb leads form the EINTHOVEN TRIANGLE



Lead II = Lead I + Lead III

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Einthoven's Triangle and Einthoven Law

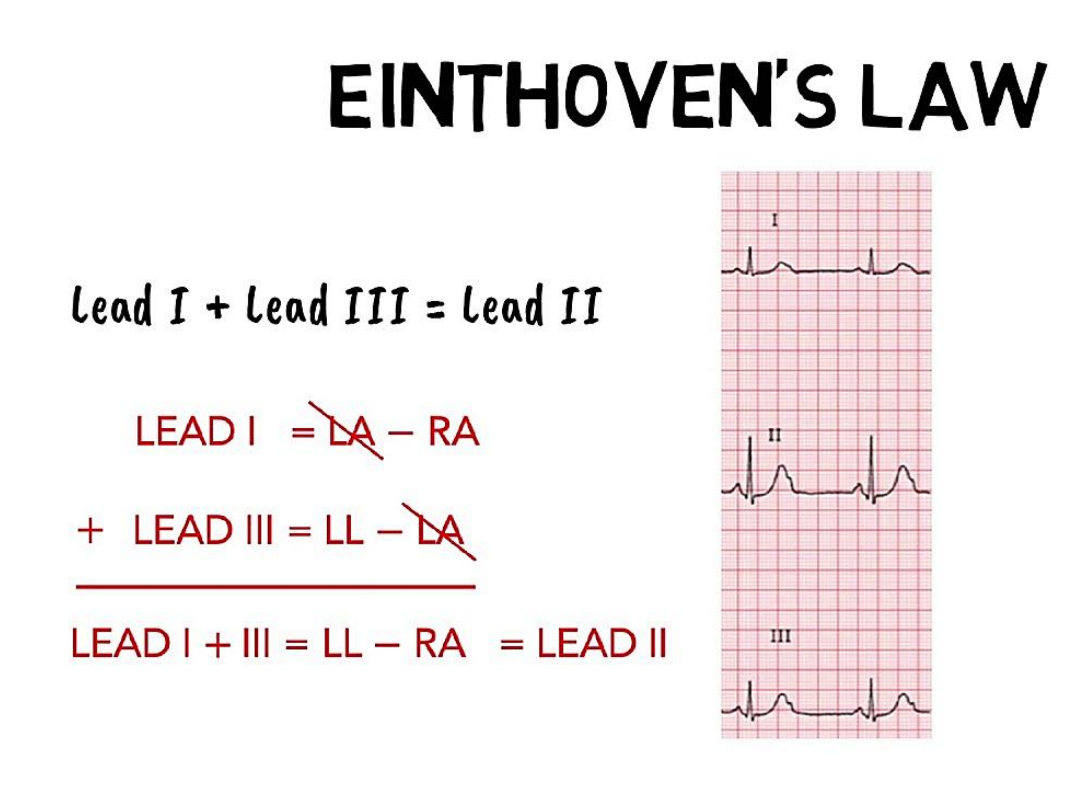
Einthoven's law Statement:

The potential of any wave in lead II is equal to the sum of the potentials in leads, I and III.

$$\boxed{\textbf{Lead II = Lead I + Lead III}}$$

Importance:

If the potentials of any 2 of the three bipolar limb leads are known, the third one can be calculated mathematically by this relationship.



Explanation of Einthoven's Law with an example

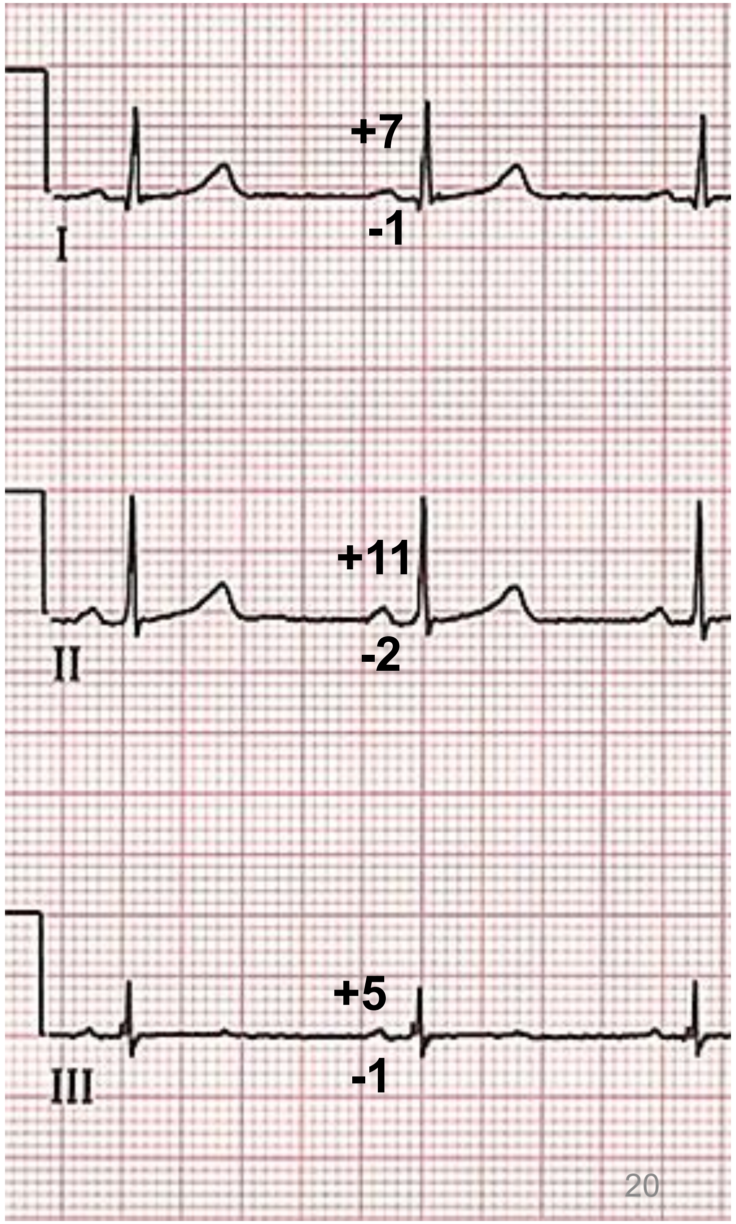
Consider QRS wave in the ECG shown in the figure:

Net amplitude of QRS in Lead I = 7-1 = 0.6 mV

Net amplitude of QRS in Lead III = 5-2 = 0.3 mV

Net amplitude of QRS in Lead II = 11-2 = 0.9 mV

Lead II (0.9) = Lead I (0.6) + Lead III (0.3)



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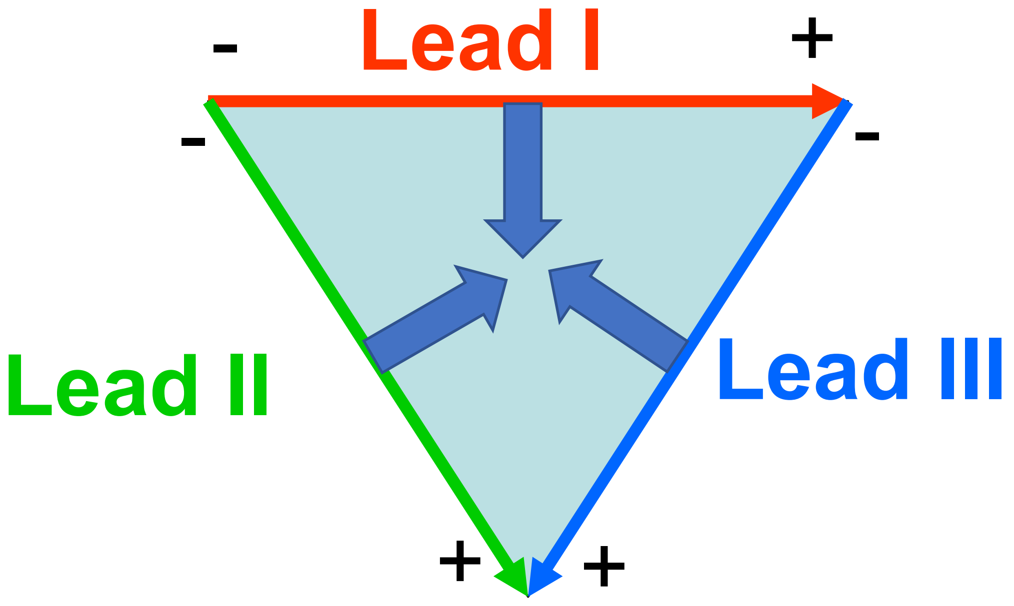
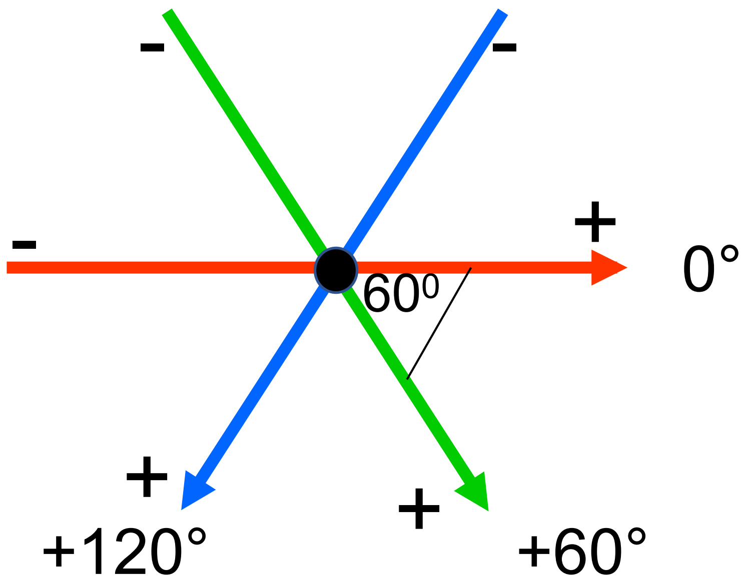
Lecture Outline

- ECG Leads and placement
- Generation of the hexaxial reference system
- Calculation of the Mean Electrical Axis & its importance
- ECG changes in cardiac arrhythmias

Generation of the hexaxial reference system

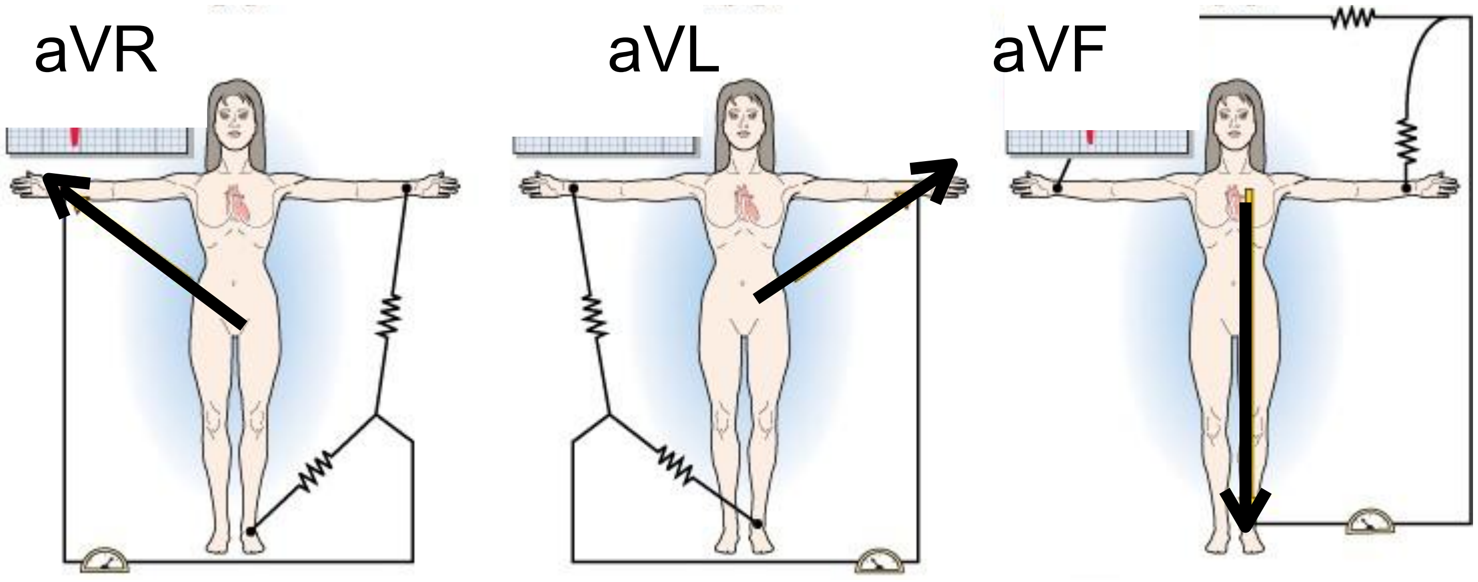
Bipolar limb lead axes projected to the center makes a triaxial reference.



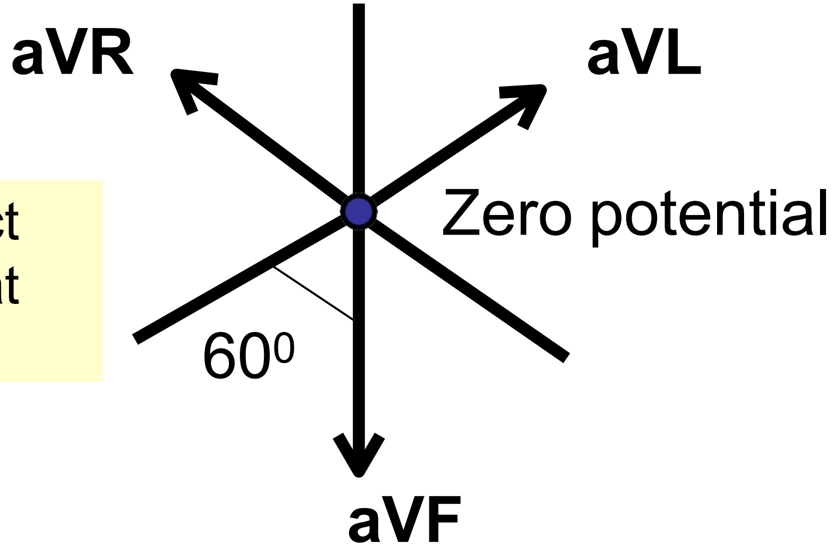


Lead I is horizontal at 0^0
Lead II is 60^0 to Lead I
Lead III is 120^0 to Lead I

Generation of the hexaxial reference system



Augmented limb lead axes projected to the center makes another triaxial reference.

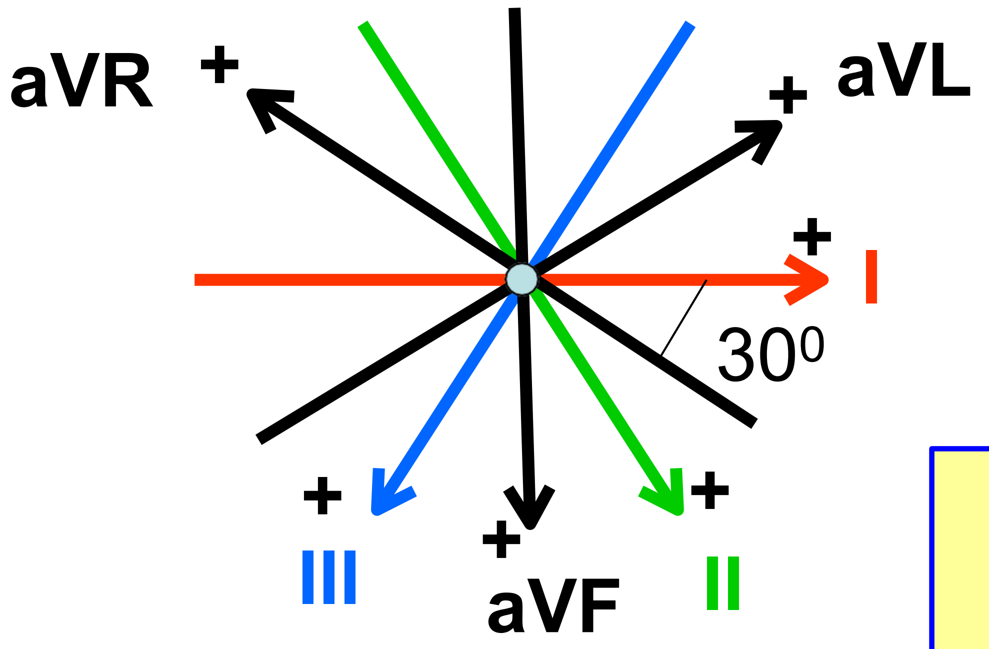


Leads bisect each other at 60^0 angles.

The 3 leads are referenced to center of heart which is at zero potential.

Generation of the hexaxial reference system

Bipolar leads superimposed on Augmented leads makes hexaxial reference system

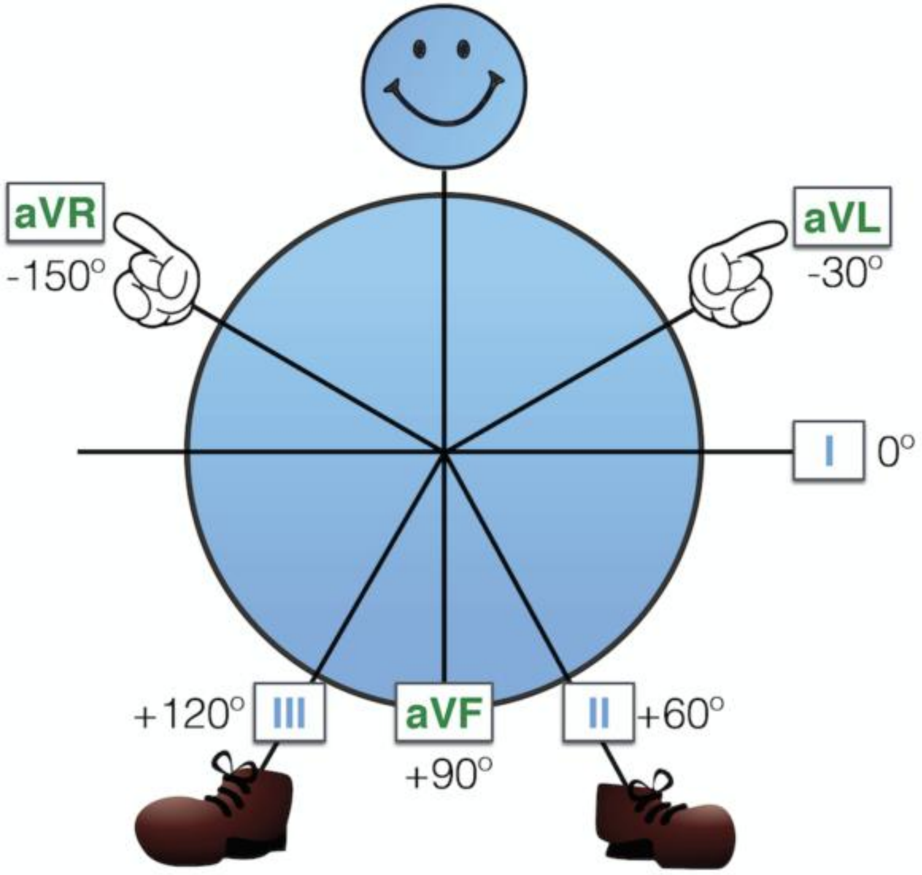


Forms the hexaxial circle

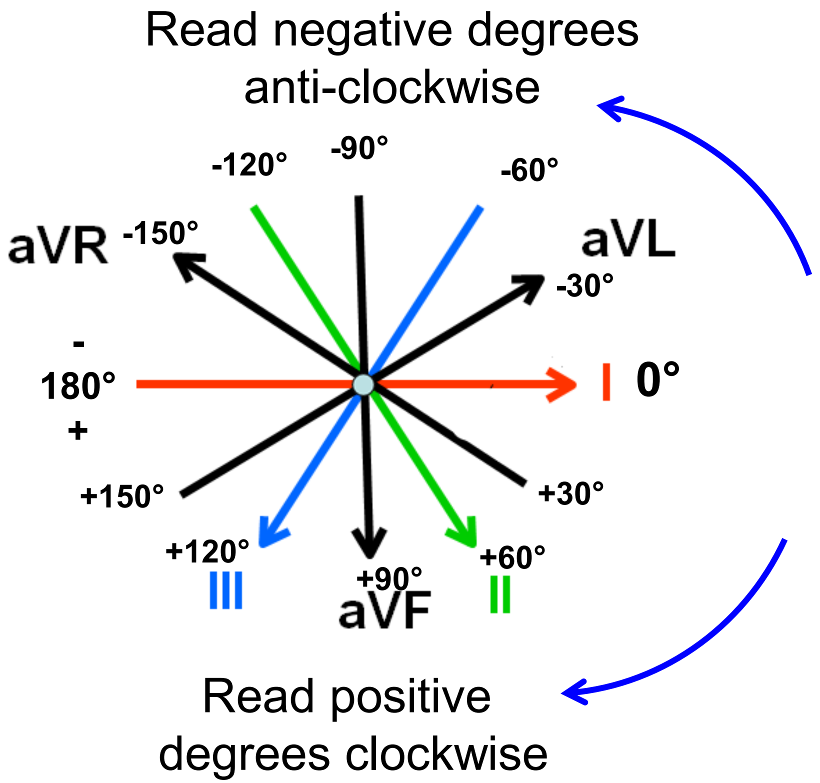
Angles are now 30^0 to each other

Note: direction of arrowhead denotes position of the Positive electrode in that Lead

Simplified way to remember the hexaxial reference system



Super Axis Man (SAM)



Note: arrowhead denotes position of the Positive electrode in that Lead

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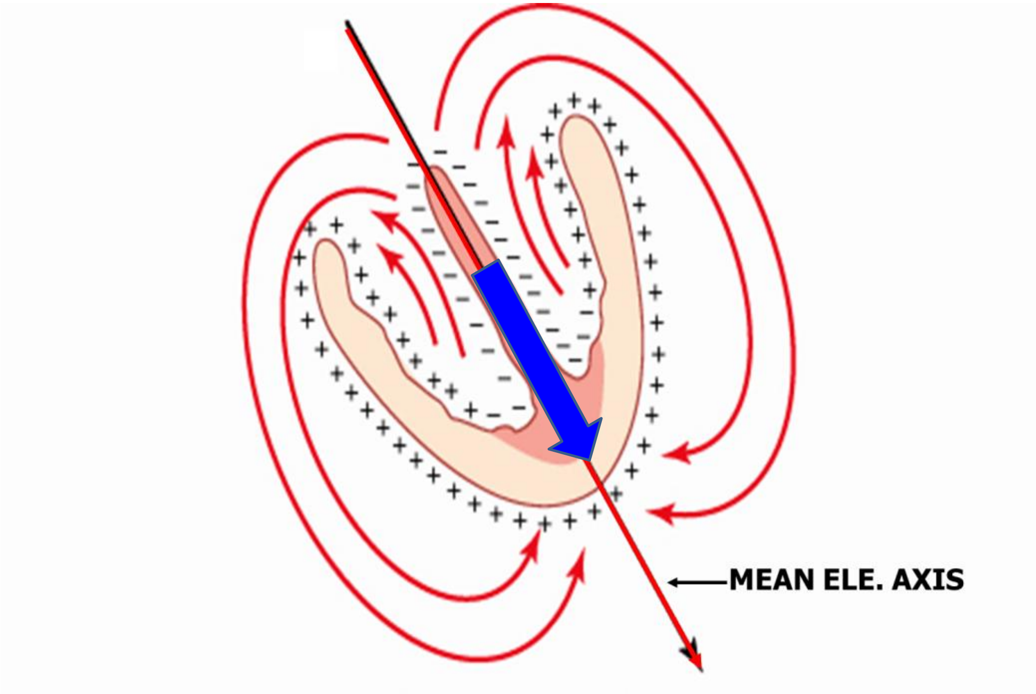
Lecture Outline

- ECG Leads and placement
- Generation of the hexaxial reference system
- Calculation of the Mean Electrical Axis & its importance
- ECG changes in cardiac arrhythmias

The Mean Electrical Axis (MEA) of the Heart

Mean **QRS vector** makes the Mean Electric Axis (MEA) of the heart.

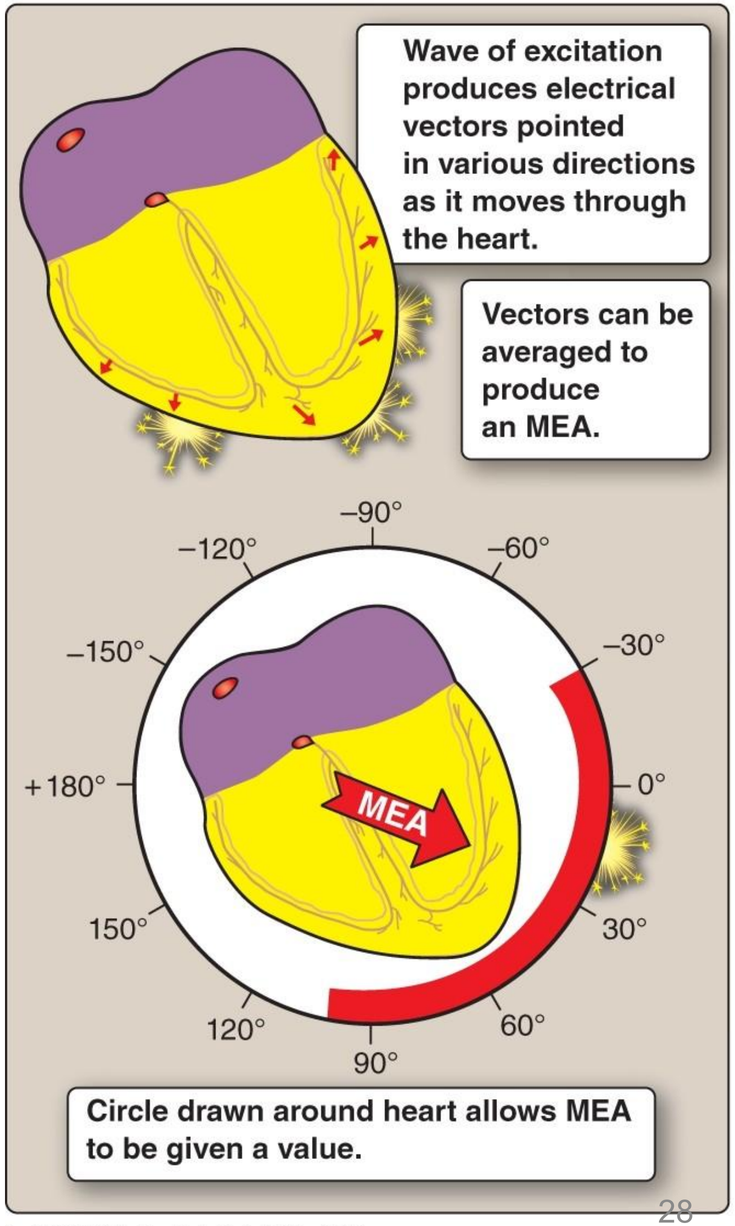


MEA is the net direction of electrical conduction during **ventricular depolarization**

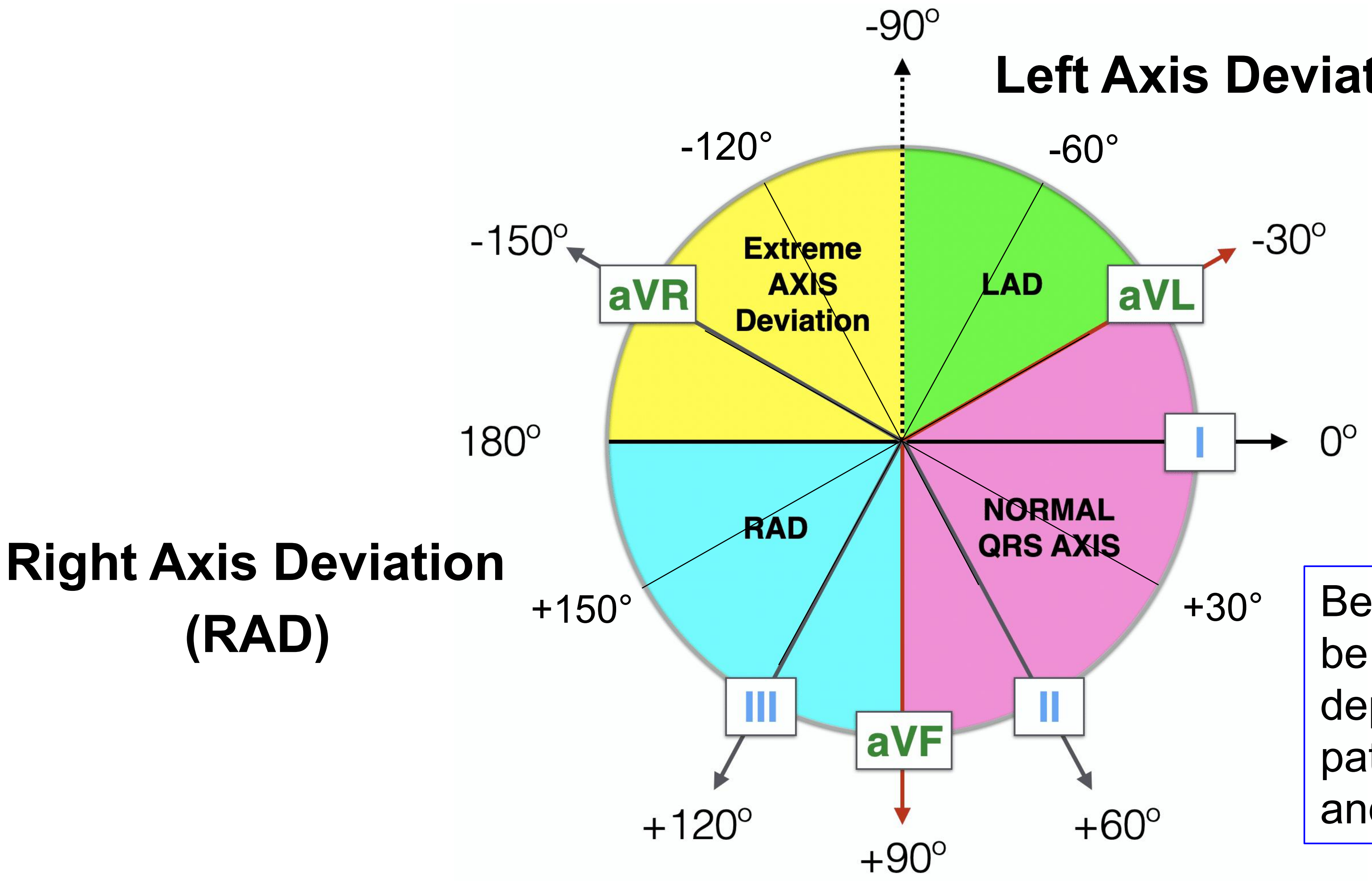
MEA can be calculated and expressed in degrees

The Mean Electrical Axis (MEA) of the Heart

- Calculated by measuring QRS complexes in the **frontal plane leads** (I, II, III, aVL, aVR, aVF)
- The precordial leads ($V_{1\text{-}6}$) are NOT used
- Ascribed a numerical value in degrees
- Abnormal MEA values indicate pathological conditions. E.g., conduction defects, ventricular hypertrophy

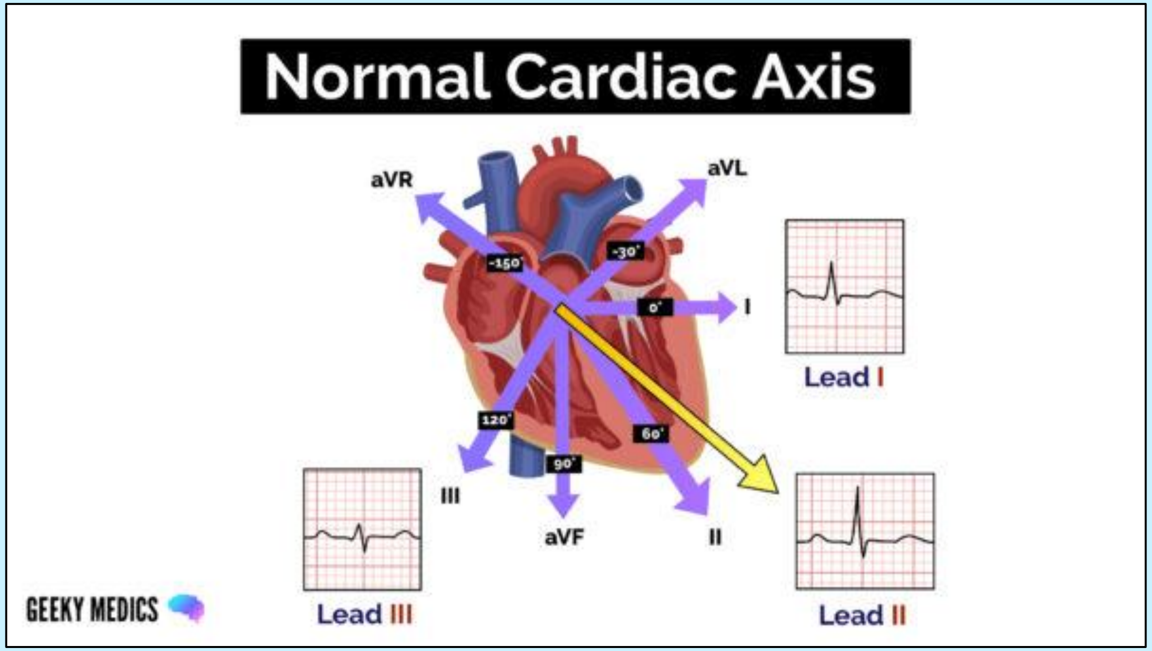


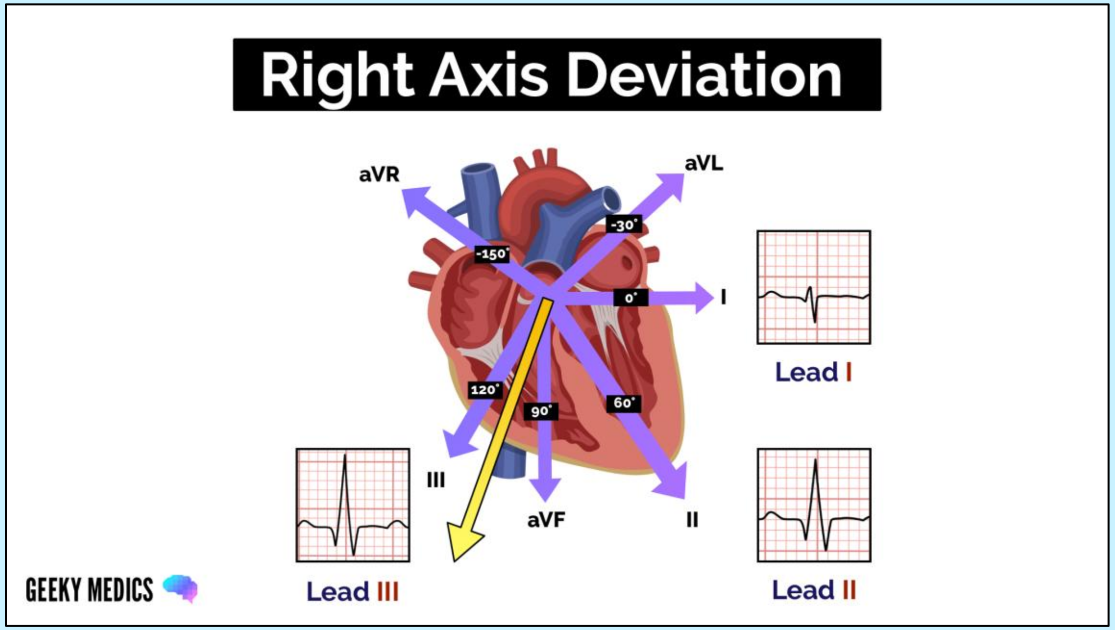
The Mean Electrical Axis (MEA) of the Heart

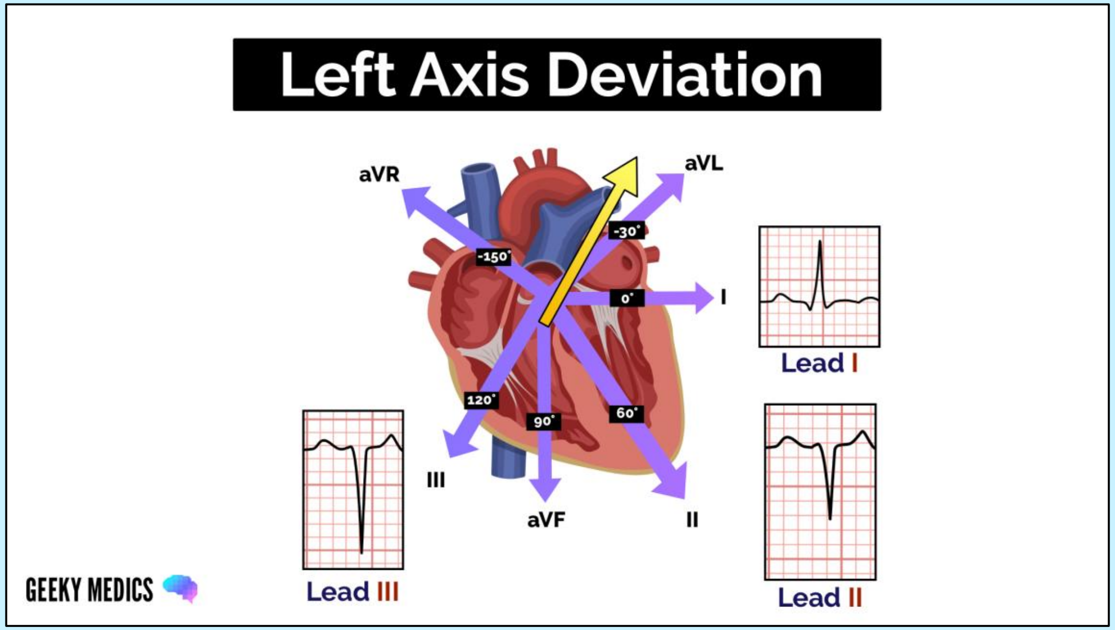


Between 0° and -30° can be normal or abnormal depending on existing pathology. MEA between 0° and 90° is **always normal**.

Comparison of normal QRS axis with Right and Left axis deviation







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Clinical Importance of MEA of the Heart

Right Axis Deviation

- Right ventricular hypertrophy
- Right bundle branch block

Left Axis Deviation

- Left ventricular hypertrophy
- Left bundle branch block

In hypertrophy, **increased muscle mass** makes the QRS vector to go in the direction of the ventricle that is thicker.
In bundle branch block, **delay in depolarization** makes the shift of the QRS vector.

How to determine MEA

- **Geometric method: Provides accurate calculation of MEA; but time-consuming.**

- **Quadrant Method**
- **Isoelectric Lead Method**
- **Two Leads Method**

Quick methods; just a look at leads!

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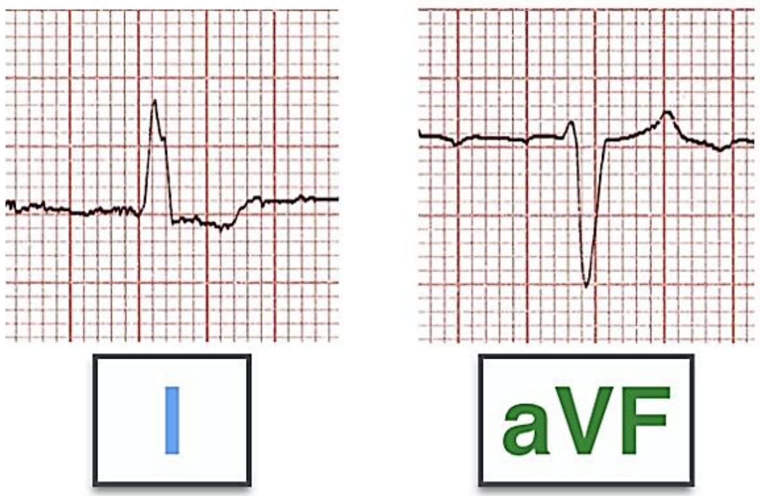
Geometric Method to determine MEA

Measure **net** amplitude of QRS in at least two of the six limb leads.

(select the largest QRS complex to minimize measuring errors)



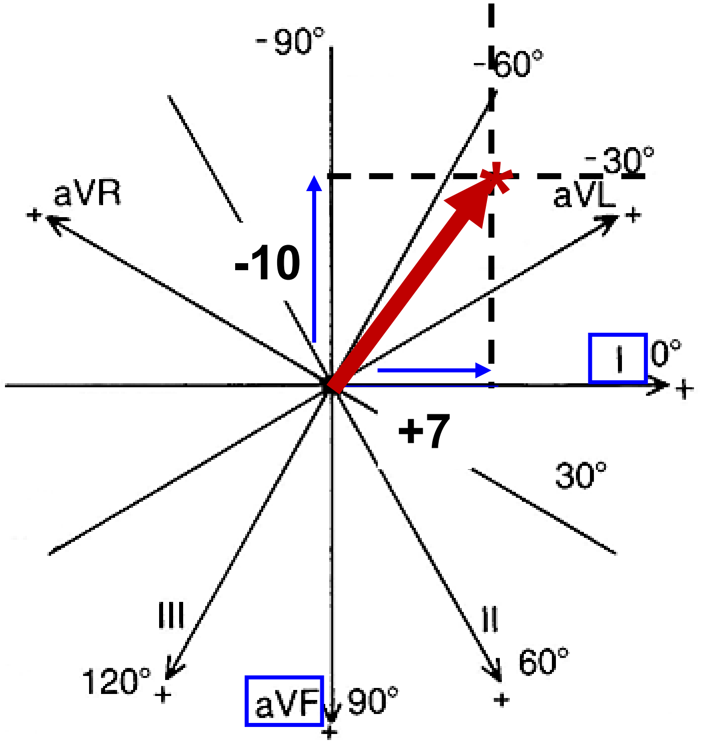
In Lead I, the deflection is 1 down and 8 up.
The NET deflection = **PLUS 7** squares.

In aVF, the deflection is 1 square up and 11 squares down.
The NET deflection is = **MINUS 10** squares.

Plot these values geometrically on the lead axis as explained in the next slide

Geometric Method to determine MEA

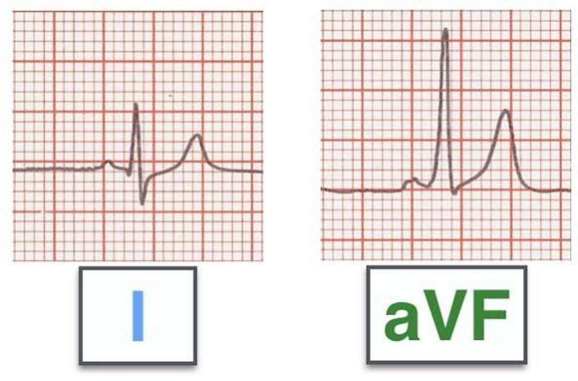
Plot +7 units on Lead I and **-10 units on negative side of aVF**. Draw perpendiculars from tip of each red arrows as shown in the figure. The point of intersection of these perpendiculars (star) indicates the highest point of the MEA.



The line joining point of intersection to the center (Red arrow) indicates MEA.

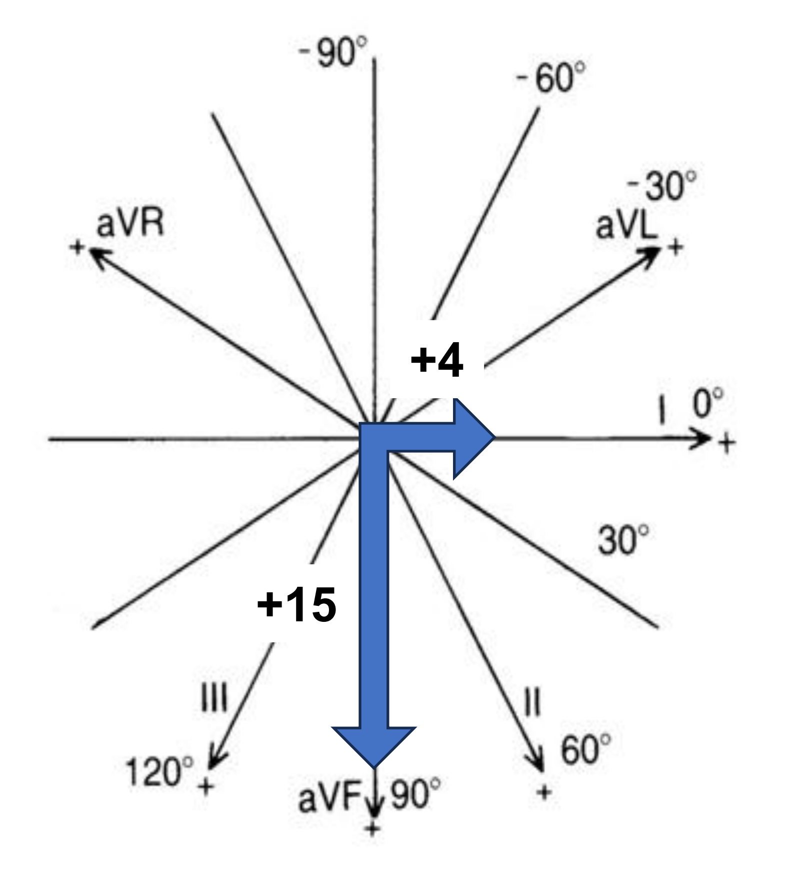
MEA in this case is between -30° and -60° and closer to -60°

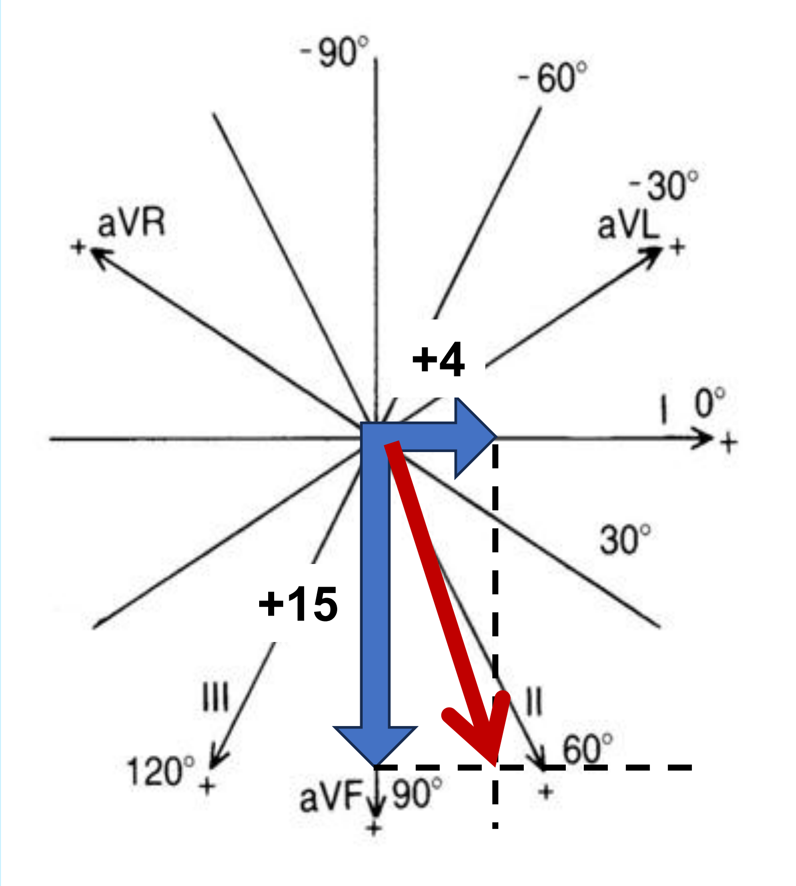
Another example of Geometric Method to determine MEA: Example



+4
(7-3=+4)

+15
(16-1=+15)





Mean Electrical Axis (Red arrow) of this heart is +75°

41

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How to determine MEA: Quick Methods

- **Quadrant Method**
- **Isoelectric Lead Method**
- **Two Leads Method**

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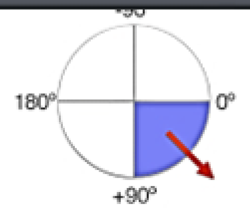
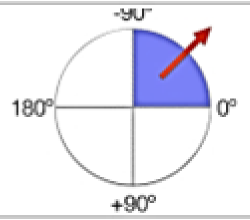
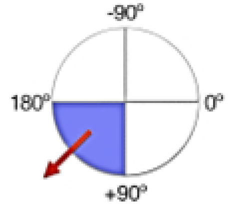
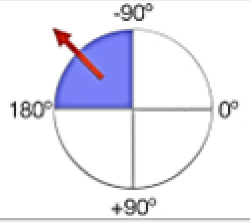
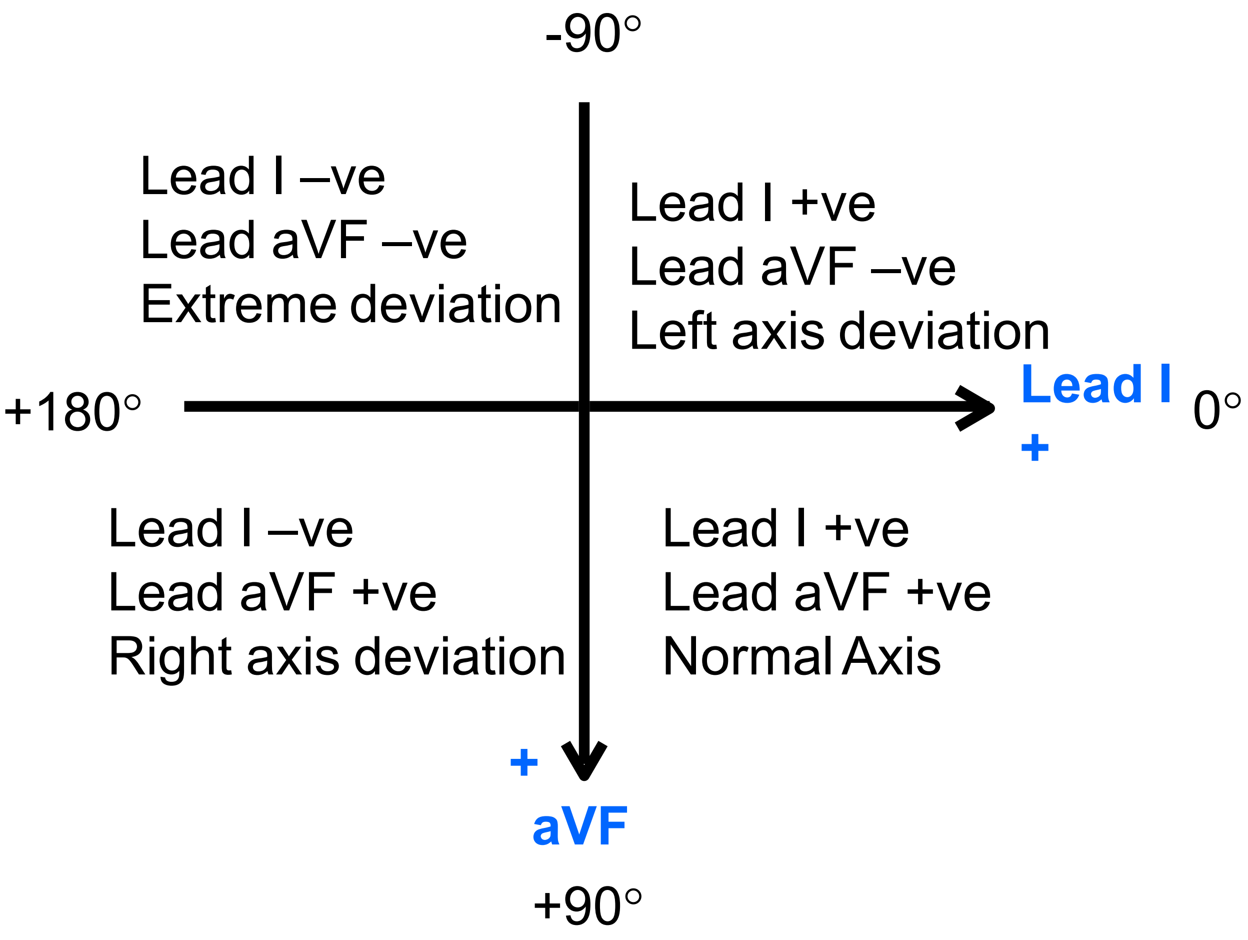
Quadrant Method to determine MEA

By just looking at the QRS waves in **Lead I and aVF**, one can tell straight away whether the MEA is normal or deviated!

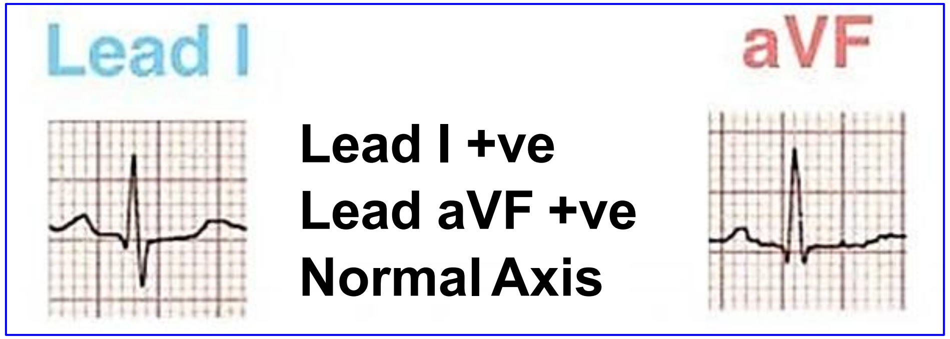
(RAD=Right Axis Deviation; LAD=Left Axis Deviation.)

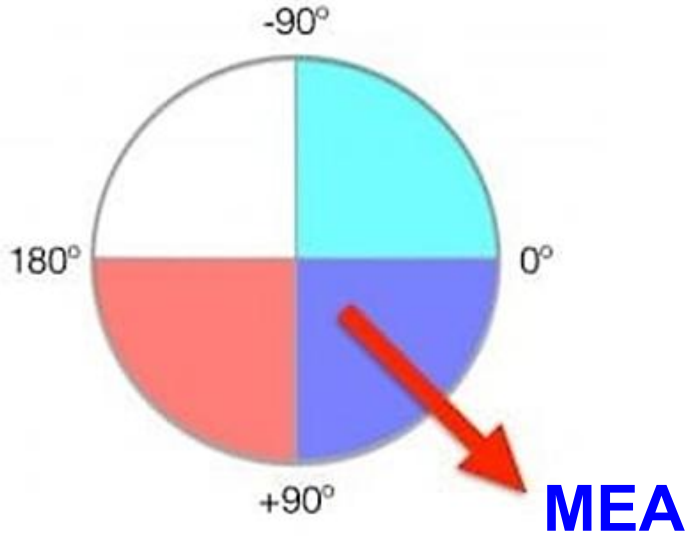
Lead 1	Lead aVF	Quadrant	Axis
POSITIVE	POSITIVE		**Normal Axis** (0 to +90°)
POSITIVE	NEGATIVE		**LAD** (0 to -90°)
NEGATIVE	POSITIVE		**RAD** (+90° to 180°)
NEGATIVE	NEGATIVE		**Extreme Axis** (-90° to 180°)

Quadrant Method to determine MEA: Example



What is the MEA of this person?





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Quadrant Method to determine MEA

The Quadrant Method A cautionary note!

A cardiac axis between 0^0 and $+90^0$ is normal.

In many instances, a cardiac axis between 0^0 and -30^0 is also normal. Some cardiologist advocate looking at the QRS complexes in Leads I and II. If they are both upright, then the axis lies between 0^0 and $+90^0$. But if the QRS complex in either Lead I or Lead II is not primarily upward then the axis is abnormal and the axis should then be determined by more accurate means (geometric method).

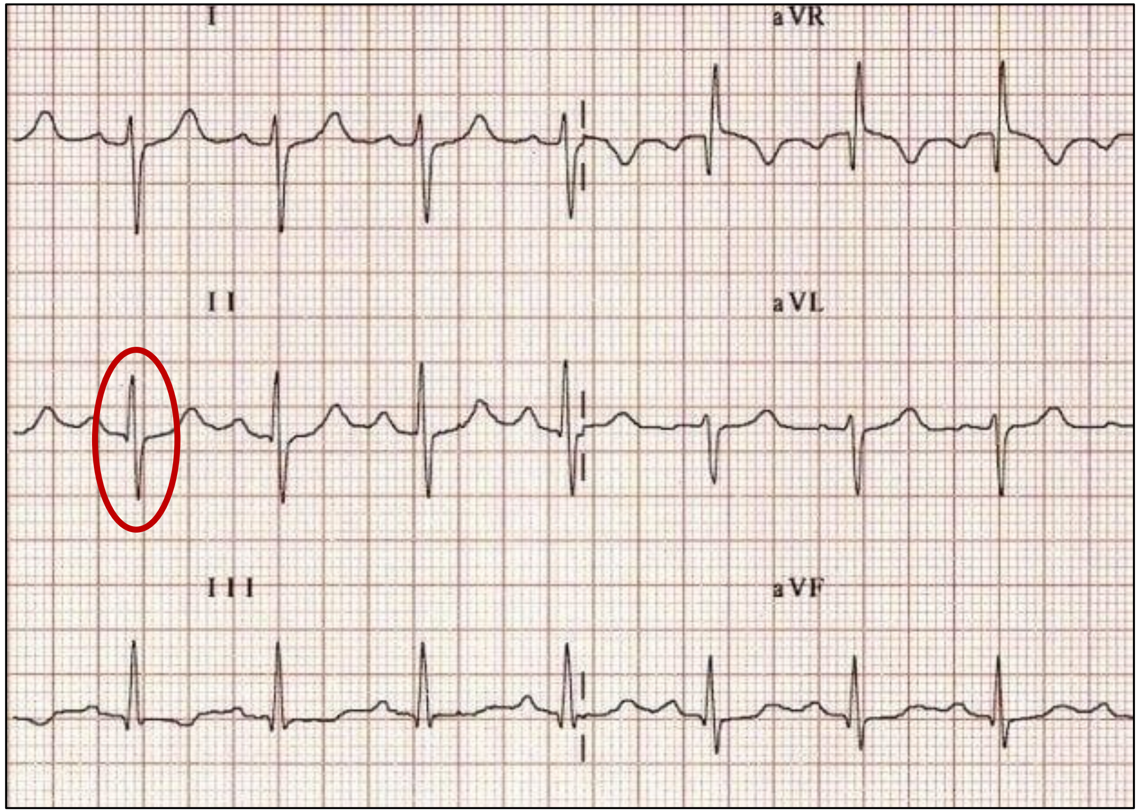
Isoelectric Lead Method to determine MEA

Inspect all 6 limb leads and find the one with the **most equiphasic (isoelectric)** QRS complex

According to the rules governing ECG waves, the lead that shows an Equiphasic QRS complex is perpendicular to the MEA of the heart.

OR

The MEA of the heart most likely lies on the lead axis that is perpendicular to the lead axis of the equiphasic QRS complex.



QRS is considered ALMOST isoelectric in Lead II in this ECG. The MEA of the heart is **perpendicular** to lead II

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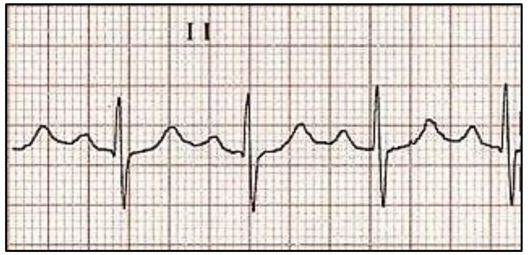
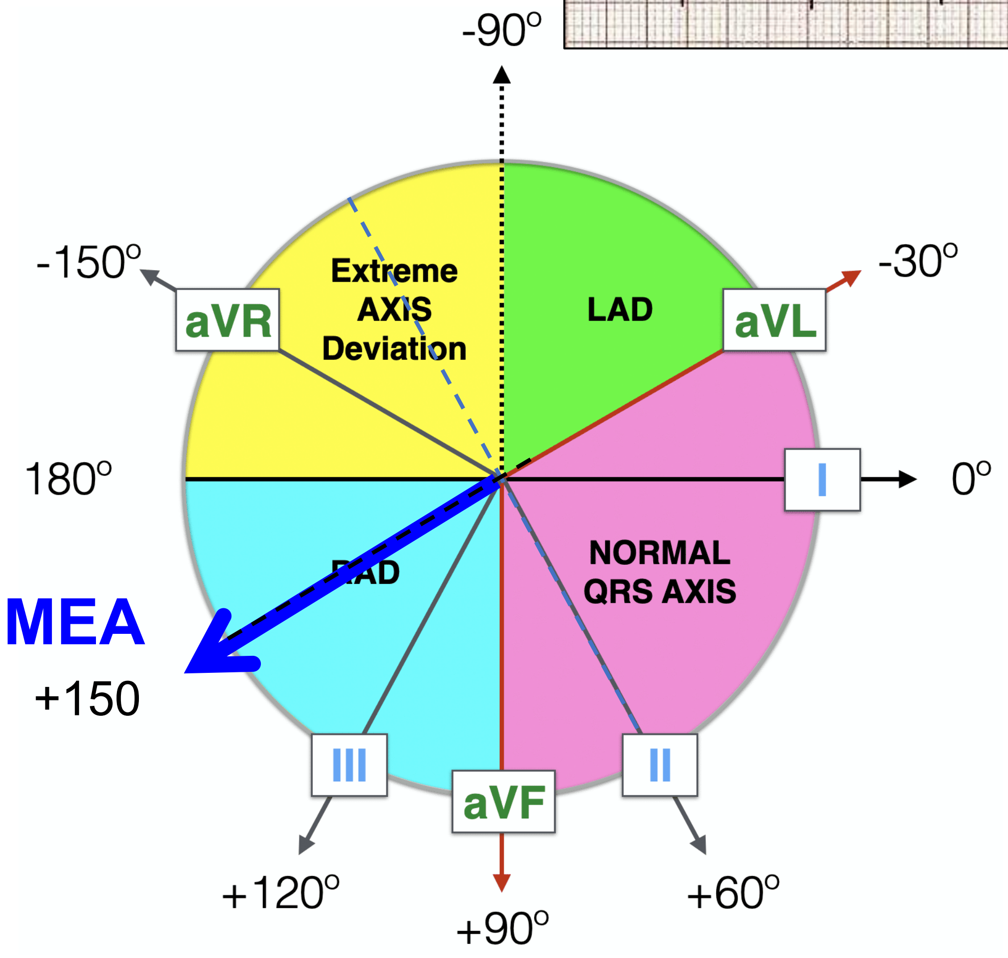
Isoelectric Lead Method to determine MEA: Example

Explanation



MEA is perpendicular to Lead II.
Therefore, MEA is on aVL axis.
But at this stage it is not known in which direction the axis is? : -30^{0} or $+150^{0}$?

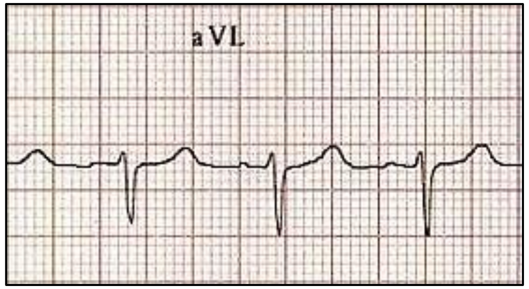


Which one is correct?

Inspect the ECG lead that is perpendicular to Lead II i.e., aVL. If QRS is +ve in aVL, then the axis points towards the +ve pole of that lead. i.e., -30^{0}.

If the QRS is negative in aVL, then the axis points towards the negative pole in aVL, i.e., $+150^{0}$.

MEA is +150



Two Leads Method to determine MEA

Select **Lead I and Lead II**; Look for Positive or Negative deflections in them

	Normal Axis (0 to +90)	LAD PATHOLOGICAL (-30 to -90)	RAD (+90 to +180)	EXTREME Axis (-90 to -180)	LAD PHYSIOLOGICAL (0 to -30)
Lead I	**POSITIVE** 	**POSITIVE** 	**NEGATIVE** 	**NEGATIVE** 	**POSITIVE** 
Lead II	**POSITIVE** 	**NEGATIVE** 	**POSITIVE** 	**NEGATIVE** 	**EQUIPHASIC**

"Thumbs leave each other"
LAD

"Thumbs reach each other"
RAD

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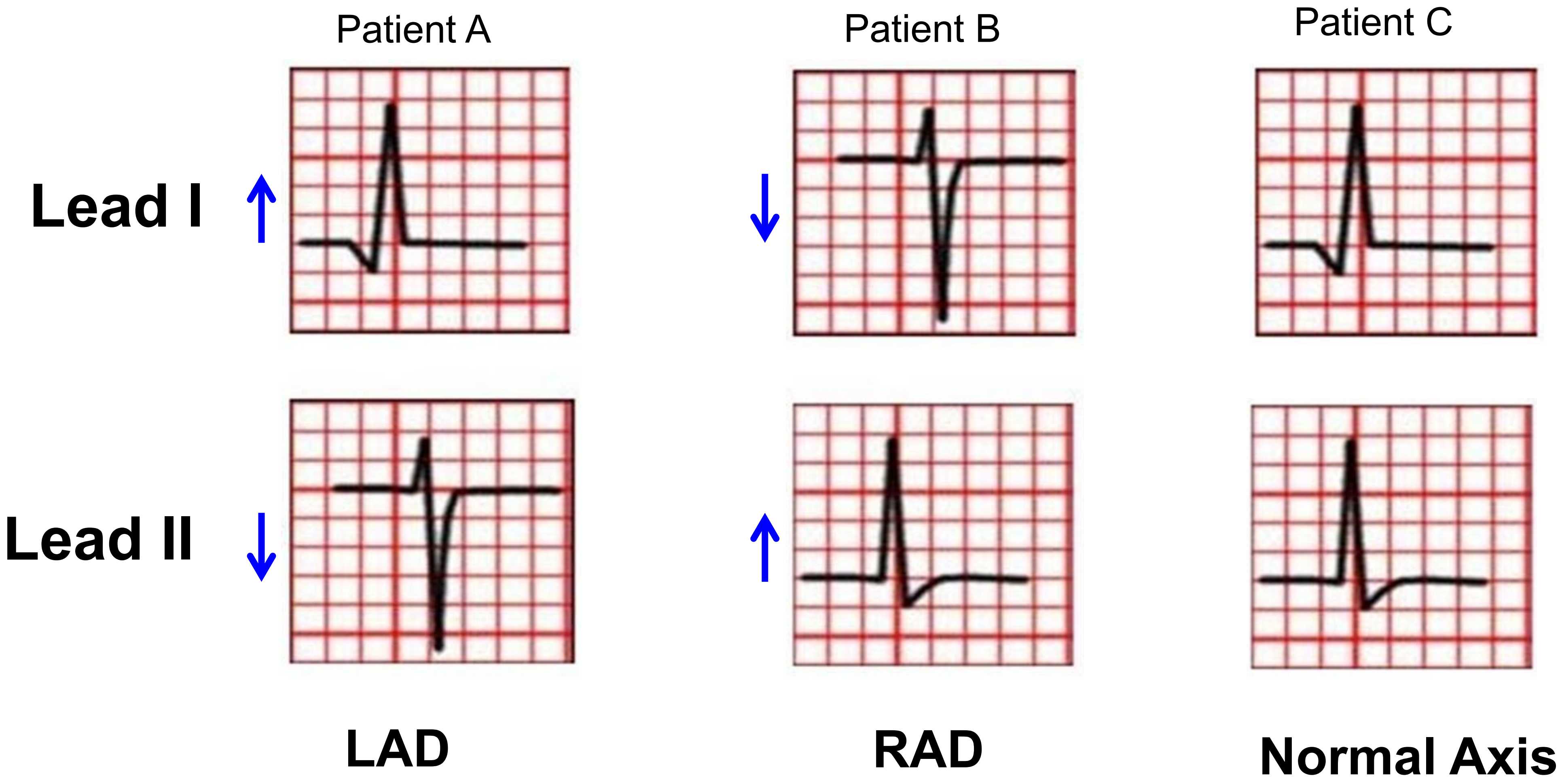
Two Leads Method to determine MEA: Examples

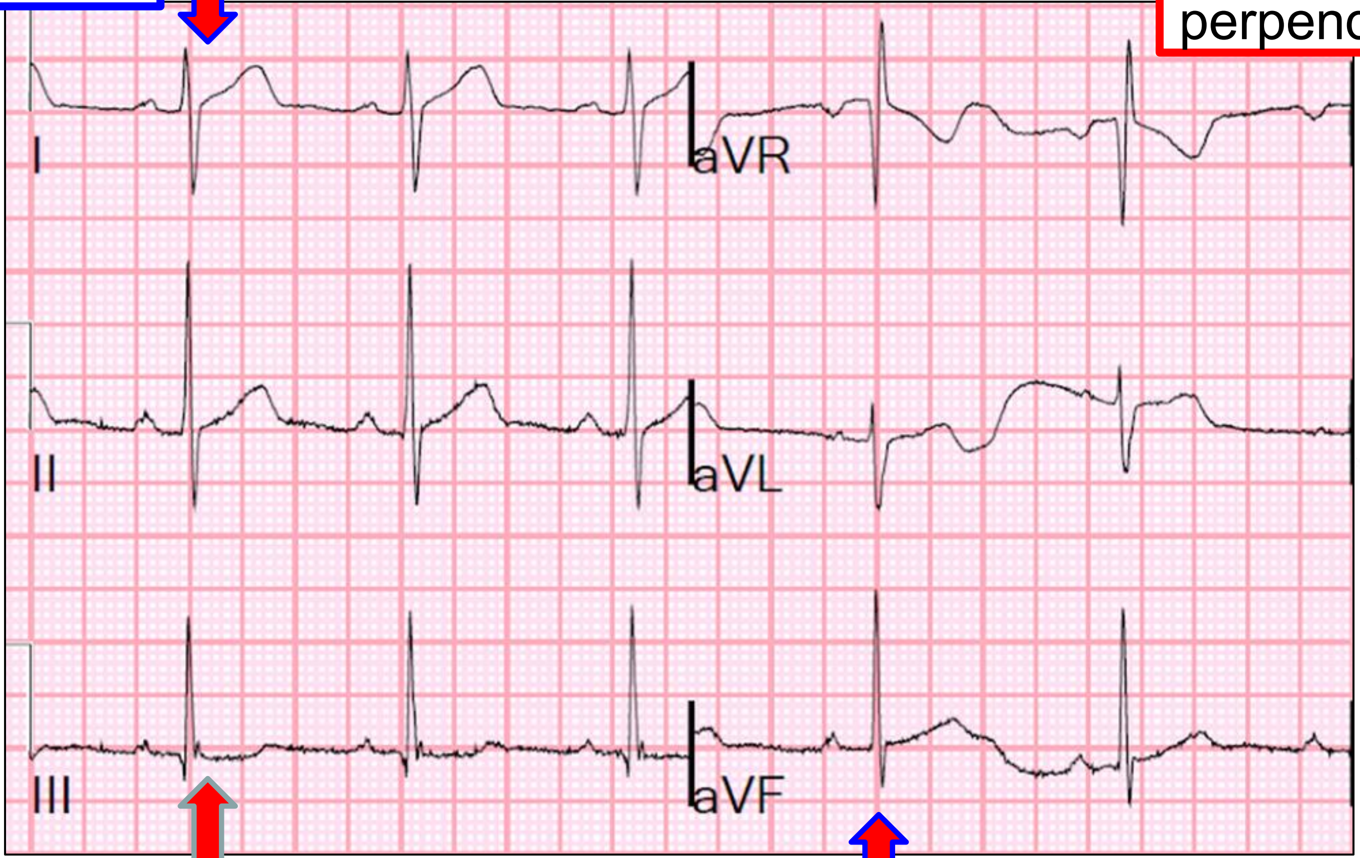


Using ECG below, determine the MEA of the heart: Apply all 3 quick methods to confirm

Negative in Lead I

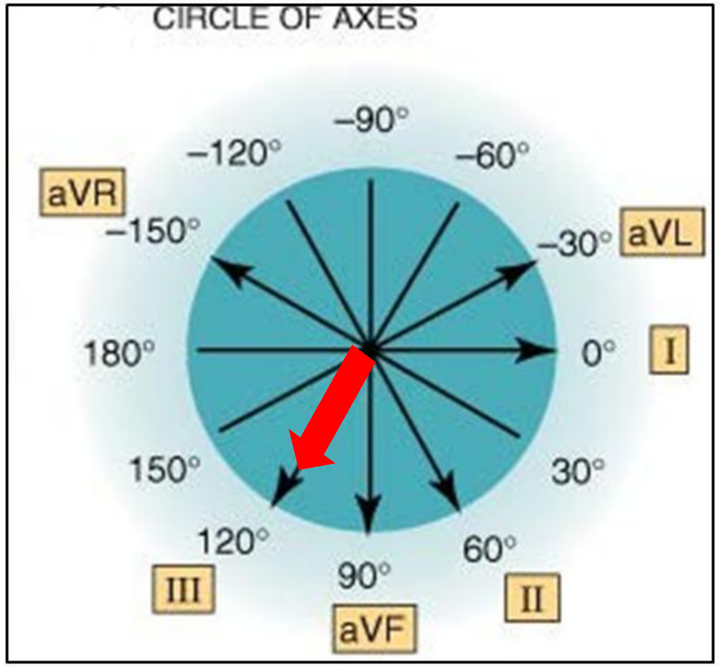
Equiphasic in Lead aVR: MEA is perpendicular to this lead axis

1. Applying Quadrant Method (blue boxes): MEA falls in lower left quadrant.



2. Applying Isoelectric Method (red boxes): MEA lies in the axis of Lead III

3. Applying 2-lead Thumb method: Right axis deviation



Lead III shows Positive QRS: MEA on positive side

Positive in Lead aVF

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Lecture Outline

- ECG Leads and placement
- Generation of the hexaxial reference system
- Calculation of the Mean Electrical Axis & its importance
- ECG changes in cardiac arrhythmias

ECG in most common arrhythmias:

- ❑ **Sinus Tachycardia**
- ❑ **Sinus Bradycardia**
- ❑ **AV conduction block (Heart block)**
- ❑ **Atrial fibrillation and**
- ❑ **Ventricular fibrillation**

There is a separate additional CPR DLA which covers details of the topic (**DLA on Applied ECG: Interpretation of abnormal ECG**). Please read this DLA after studying lectures on ECG.

ECG in Sinus Tachycardia

- Everything is NORMAL in ECG except the **heart rate is more than 100/min**.

- What is the heart rate in the above ECG?
- Heart rate = 300/2.4 large boxes = 125/min.

- Sinus tachycardia occurs in sympathetic stimulation, hyperthyroidism, fever, and treatment with beta agonists.

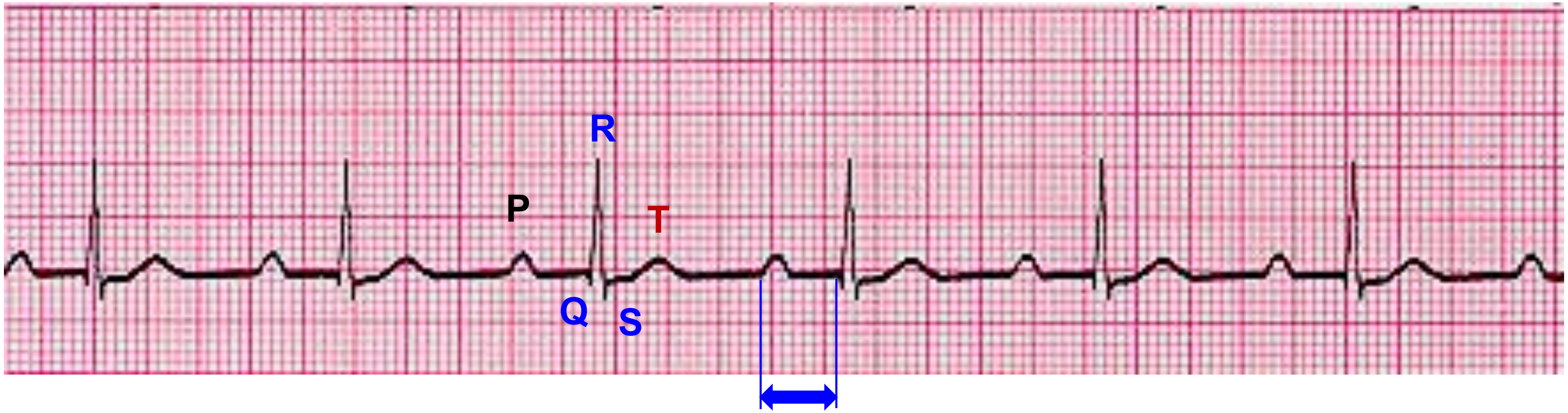
ECG in Sinus Bradycardia

- Everything is NORMAL in ECG except the **heart rate is less than 60/min**.

- What is the heart rate in the above ECG?
- Heart rate = 300/5.7 large boxes = 52/min.

- Sinus bradycardia occurs in vagus nerve stimulation, carotid massage, hypothyroidism, and treatment with beta blockers.

ECG in AV conduction block (Heart block)



- Heart rate may be normal or slow (In the above ECG, HR is = 300/4.8 = 62/min).
- All ECG waves are normal.
- PR interval is **more than 5 small boxes**.
- PR interval is prolonged in AV conduction blocks.
- AV nodal conduction is delayed in heart block.
- There are 3 degrees (severities) of heart blocks.

In the above ECG, PR interval = 7 small boxes
= 7x0.04
= 0.28 sec.
(Normal PR int= 0.12 to 0.20 sec)

ECG in AV conduction block (Heart block)

- There are 3 severities of heart blocks:
- First degree, second degree & third-degree blocks.

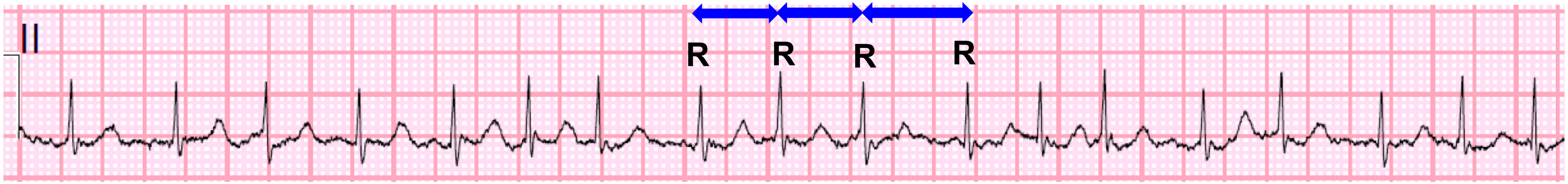
- First degree heart block: PR is longer than normal but all impulses arriving at AV node pass to the ventricles.

- Second degree block: PR interval progressively prolong in successive beat until some impulse fails to pass through the AV node to the ventricles.

- Third degree heart block: Absolutely NO impulses pass through the AV node to reach ventricles. This is called complete AV block.

ECG in Atrial Fibrillation

Multiple sites generate electrical impulses in atria. There is no single wave of depolarization spreading over the atria.



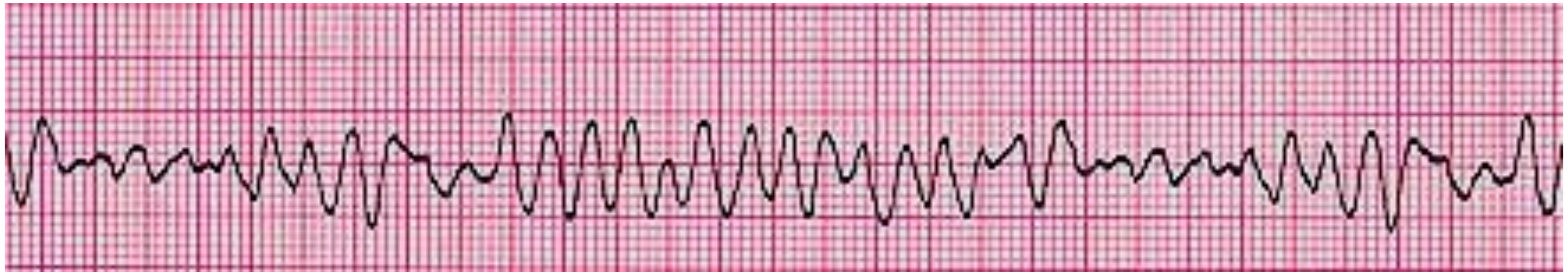
ECG findings:
i) P waves are absent in Atrial fibrillation.

ii) **Irregular R-R intervals**. Due to refractory period of AV node only some signals pass through AV node to ventricles at irregular intervals. This produces an irregular heart rhythm.

The atria do not contract;
There is no atrial systole;
The last part of ventricular filling from atrial kick is absent!

ECG in Ventricular Fibrillation (VF)

Multiple sites generate electrical impulses in the ventricles. There is no single wave of depolarization spreading over the ventricles.



Defibrillation required!

The ECG in VF shows rapid (300 to 400/min), **irregular, shapeless** QRST **undulations** of variable amplitude, morphology, and interval.

Ventricles do not contract; therefore, **NO cardiac output and NO recordable BP!** Patients invariably present with sudden collapse (syncope and/or sudden cardiac arrest).

Any Questions ?

How to reach me: Either walk into my office or email.

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Other References

1. Goldberger's Clinical Electrocardiography, A simplified approach. Ninth Edition. Goldberger, Ary L., MD, FACC. 2018 Elsevier Inc.

2. LIFE IN THE FASTLANE https://litfl.com/ecg-library/